



Cochrane
Library

Cochrane Database of Systematic Reviews

Ultrasound guidance versus anatomical landmarks for neuraxial anaesthesia in adults (Review)

Makino Y, Miyake K, Roche D, Yoshimura S, Nahara I, Sahker E, Watanabe N

Makino Y, Miyake K, Roche D, Yoshimura S, Nahara I, Sahker E, Watanabe N.
Ultrasound guidance versus anatomical landmarks for neuraxial anaesthesia in adults.
Cochrane Database of Systematic Reviews 2025, Issue 5. Art. No.: CD014964.
DOI: [10.1002/14651858.CD014964.pub2](https://doi.org/10.1002/14651858.CD014964.pub2).

www.cochranelibrary.com

Ultrasound guidance versus anatomical landmarks for neuraxial anaesthesia in adults (Review)

Copyright © 2025 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

WILEY

TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	2
SUMMARY OF FINDINGS	4
BACKGROUND	6
OBJECTIVES	7
METHODS	7
RESULTS	10
Figure 1.	11
Figure 2.	15
Figure 3.	17
Figure 4.	18
Figure 5.	20
Figure 6.	21
Figure 7.	22
Figure 8.	23
Figure 9.	24
DISCUSSION	25
AUTHORS' CONCLUSIONS	26
SUPPLEMENTARY MATERIALS	26
ADDITIONAL INFORMATION	26
REFERENCES	28
ADDITIONAL TABLES	32
INDEX TERMS	48

[Intervention Review]

Ultrasound guidance versus anatomical landmarks for neuraxial anaesthesia in adults

Yuto Makino¹, Kentaro Miyake², David Roche^{3,4}, Satoshi Yoshimura⁵, Isao Nahara⁶, Ethan Sahker^{7,8}, Norio Watanabe⁹

¹Department of Anesthesiology and Intensive Care Medicine, Nagoya City University Graduate School of Medical Sciences, Nagoya, Japan. ²Department of Pharmacoepidemiology, School of Public Health in the Graduate School of Medicine, Kyoto University, Kyoto, Japan. ³Evidence Synthesis Ireland, Galway, Ireland. ⁴Department of Anaesthesia and Pain Management, The Royal Children's Hospital, Melbourne, Australia. ⁵Department of Preventive Services, Kyoto University Graduate School of Medicine/School of Public Health, Kyoto, Japan. ⁶Department of Anesthesiology and Critical Care Medicine, Kasugai Municipal Hospital, Kasugai, Japan. ⁷Department of Health Promotion and Human Behavior, Graduate School of Medicine/School of Public Health, Kyoto University, Kyoto, Japan. ⁸Population Health and Policy Research Unit, Medical Education Center, Graduate School of Medicine, Kyoto University, Kyoto, Japan. ⁹Department of Psychiatry, Soseikai General Hospital, Kyoto, Japan

Contact: Yuto Makino, yuuto.m78@gmail.com.

Editorial group: Cochrane Central Editorial Service.

Publication status and date: New, published in Issue 5, 2025.

Citation: Makino Y, Miyake K, Roche D, Yoshimura S, Nahara I, Sahker E, Watanabe N. Ultrasound guidance versus anatomical landmarks for neuraxial anaesthesia in adults. *Cochrane Database of Systematic Reviews* 2025, Issue 5. Art. No.: CD014964. DOI: [10.1002/14651858.CD014964.pub2](https://doi.org/10.1002/14651858.CD014964.pub2).

Copyright © 2025 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.

ABSTRACT

Rationale

Neuraxial anaesthesia can be difficult to administer successfully, because the targeting space imagined by palpation of anatomical position can deviate from the actual position. Successful neuraxial anaesthesia, with as few punctures as possible, is important to reduce complications and increase patient satisfaction. Neuraxial anaesthesia by ultrasound guidance may be a useful way to increase success rates.

Objectives

To assess the clinical efficacy and safety of ultrasound guidance compared with anatomical landmarks for neuraxial anaesthesia in adults.

Search methods

We searched CENTRAL, MEDLINE, Embase, Web of Science, and two trials registries, together with reference checking and citation searching, to identify studies that are included in the review. After the original search on 11 October 2022, we updated the electronic searches on 28 November 2023.

Eligibility criteria

We included randomised controlled trials (RCTs) that compared ultrasound guidance with the use of conventional anatomical landmarks for neuraxial anaesthesia in adults (≥ 18 years). We excluded studies on non-anaesthetic neuraxial procedures, such as lumbar puncture for diagnosis.

Outcomes

Our critical outcomes were: (1) number of attempts until success or procedure termination; (2) procedure time; and (3) participant satisfaction during the procedure. Our important outcomes were: (1) first-attempt success; (2) technical failure; (3) pain during the procedure; and (4) any adverse events.

Risk of bias

We used the Cochrane RoB 2 tool to assess risk of bias in the included studies for the seven critical and important outcomes.

Synthesis methods

We synthesised results for each outcome, where possible, by using a random-effects analytical model. We calculated risk ratios (RR) for dichotomous outcomes, and mean differences (MD) or standardised mean difference (SMD) for continuous outcomes, each with 95% confidence intervals (CI). Where meta-analysis was not possible due to the nature of the data, we summarised the results narratively. We used GRADE to assess the certainty of evidence for each outcome.

Included studies

We included a total of 65 studies with 6823 participants. The studies were conducted in Africa, Asia, Europe, the Middle East, North America, and Oceania, and were published between 2001 and 2023. Thirty-two studies evaluated obstetric populations undergoing labour epidural or caesarean section (3149 participants); the remaining studies evaluated participants who received surgery under neuraxial anaesthesia. Thirty-three studies evaluated spinal anaesthesia, 15 studies evaluated lumbar epidural anaesthesia, 12 studies evaluated combined spinal and epidural anaesthesia, and the remaining four studies evaluated thoracic epidural anaesthesia. Four studies evaluated real-time ultrasound as an intervention arm. Two studies were three-arm trials, which evaluated pre-procedural ultrasound and real-time ultrasound. For participant satisfaction, a variety of measurement scales were used, so we synthesised participant satisfaction using SMD.

Synthesis of results

Our meta-analyses demonstrated that compared with anatomical landmarks, ultrasound guidance reduces the number of attempts until success or procedure termination (MD -0.41 attempts, 95% CI -0.51 to -0.31; 57 studies, 6192 participants; high-certainty evidence), and procedure (needling) time (MD -33.8 seconds, 95% CI -47.22 to -20.39; 43 studies, 4178 participants; high-certainty evidence). Ultrasound guidance likely increases the rate of first-attempt success (RR 1.40, 95% CI 1.29 to 1.52; 50 studies, 5205 participants; moderate-certainty evidence). Ultrasound guidance may result in little to no difference in participant satisfaction (SMD 0.16, 95% CI -0.03 to 0.35; 25 studies; low-certainty evidence), or in technical failure (RR 0.89, 95% CI 0.62 to 1.29; 40 studies, 4581 participants; low-certainty evidence). The evidence is very uncertain about the effects of ultrasound guidance on pain during the procedure (MD -0.20 points, 95% CI -0.52 to 0.13; 13 studies, 1532 participants; very low-certainty evidence), and adverse effects (RR 0.71, 95% CI 0.48 to 1.03; 15 studies, 1692 participants; very low-certainty evidence).

Authors' conclusions

Compared to anatomical landmarks, ultrasound guidance for neuraxial anaesthesia in adults reduces the number of attempts required for success and reduces procedure (needling) time. It likely increases the rate of first-attempt success. Low-certainty evidence suggests that ultrasound guidance may result in little to no difference in participant satisfaction or technical failure. The evidence is very uncertain about the effect of ultrasound guidance on pain and adverse events. Although ultrasound guidance can be beneficial for neuraxial anaesthesia without increased risk, these results should be interpreted with caution due to some uncertainties in the evidence.

Funding

None.

Registration

Protocol available via doi.org/10.1002/14651858.CD014964.

PLAIN LANGUAGE SUMMARY

Is it better to use ultrasound or anatomical landmarks to guide the needle to the correct placement for anaesthesia around the spinal cord in adults?

Key messages

- Compared to anatomical landmarks, using ultrasound guidance for neuraxial anaesthesia in adults reduces the number of attempts until success, and the procedure (needling) time. Success is defined as either the number of times the needle is advanced following a backward movement, or the number of times it punctures the skin.
- Ultrasound guidance likely increases the rate of success on the first try.
- Ultrasound guidance may make little to no difference to participant satisfaction or technical failure. Technical failure might include the need to change to general anaesthesia, or the need to discontinue the procedure for neuraxial anaesthesia, or whether the person experienced the intended anaesthetic effect.
- We do not know whether ultrasound guidance affects pain during the procedures and adverse events.
- Ultrasound guidance in neuraxial anaesthesia probably has some benefits, but uncertainties remain.

What is neuraxial anaesthesia?

Neuraxial anaesthesia is performed by injecting a local anaesthetic into the spaces surrounding the spinal cord, usually the spinal or epidural space, or a combination. It is a common technique for numbing a specific area of the body, used mainly for surgery or during labour. In neuraxial anaesthesia, the doctor places a needle between the vertebrae (backbones). Once the doctor finds the right spot, he or she will either inject the anaesthetic directly, or place a thin tube, called a catheter, into the space, through which they can inject the anaesthetic. Successful neuraxial anaesthesia, with as few attempts as possible, is important to reduce complications.

What is ultrasound and how it might work?

Ultrasound uses high-frequency sound waves to create images of the inside of the body. The doctor uses a handheld device, called a probe, to send ultrasound waves into the body and create pictures. Conventional neuraxial anaesthesia is performed by using specific anatomical points on the body as a guide. Sometimes the needle does not reach the targeted space, because the anatomical position the doctor thought he or she found by touching the area can be off a bit from the actual position. Ultrasound allows the doctor to see the local anatomy of interest and the size of important structures, and helps to figure out how deep and in what direction to insert the needle.

Two ultrasound-guided methods are commonly used. In the first one, ultrasound is used before the needle is inserted, or pre-procedural. The second one is referred to as real-time, which means the needle is inserted first, then moved to the correct spot while watching the ultrasound images.

What did we want to find out?

We wanted to find out whether it was better to use ultrasound or anatomical landmarks to guide the needle for neuraxial anaesthesia in adults. We also wanted to find out whether ultrasound guidance led to similar unwanted effects as anatomical landmarks.

What did we do?

We searched for studies that investigated ultrasound guidance compared with anatomical landmarks for neuraxial anaesthesia in adults. We compared and summarised the results of the studies, and rated our confidence in the evidence, based on factors, such as study methods and sizes.

What did we find?

We found 65 studies with 6823 people. Studies were conducted in Africa, Asia, Europe, the Middle East, North America, and Oceania. Four studies evaluated real-time ultrasound as an intervention.

- Compared to anatomical landmarks, ultrasound guidance reduces the number of attempts required for successful placement of the needle by an average of 0.41 attempts per person, and reduces the procedure (needling) time by an average of 34 seconds.
- Ultrasound guidance likely increases the success rate for first attempts: 778 people out of 1000 experienced successful first attempts with ultrasound guidance, compared to 556 people out of 1000 with anatomical landmarks.
- Ultrasound guidance may result in little to no difference in participant satisfaction or technical failure.
- Ultrasound guidance may have little to no effect on pain during procedures or unwanted effects, but we are very uncertain about the results.

What are the limitations of the evidence?

We are moderately to highly confident in our findings regarding the number of attempts, the length of time for the procedure, and the rate of first-attempt success.

We have little confidence in the evidence for:

- Participant satisfaction, because it was difficult to hide who received ultrasound guidance. The results were very inconsistent across the different studies.
- Technical failure, because there were not enough studies to be certain about the results of this outcome.
- Pain during the procedure, because it was difficult to hide who received ultrasound guidance. The results were very inconsistent across the different studies, and the results included both those who experienced pain and those who did not.
- Unwanted effects, because it was difficult to hide who received ultrasound guidance, and there were not enough studies to be certain about the results. This outcome should also be interpreted with caution, because there were a variety of definitions in each study, such as back pain or headache.

How up to date is this evidence?

The evidence is up-to-date to November 2023.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Ultrasound guidance compared to anatomical landmarks for neuraxial anaesthesia in adults

Ultrasound guidance compared to anatomical landmarks for neuraxial anaesthesia in adults

Patient or population: neuraxial anaesthesia in adults
Setting: various surgical procedures in operating rooms/labour room
Intervention: ultrasound guidance
Comparison: anatomical landmarks

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	N° of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with anatomical landmarks	Risk with ultrasound guidance				
Number of attempts until success or procedure termination	The mean number of attempts until success or procedure termination was 1.96	MD 0.41 lower (0.51 lower to 0.31 lower)	-	6192 (57 RCTs)	⊕⊕⊕⊕ High ^{a,b}	
Procedure time (needling time) assessed with: seconds	The mean procedure time (needling time) was 200.8	MD 33.8 lower (47.22 lower to 20.39 lower)	-	4178 (43 RCTs)	⊕⊕⊕⊕ High ^{c,d}	
Participant satisfaction during the procedure ^e	-	SMD 0.16 higher (0.03 lower to 0.35 higher)	-	(25 RCTs)	⊕⊕⊕⊖ Low ^{f,g}	We interpreted the results of SMD as follows: 0.2 indicates a small effect, 0.5 indicates a medium effect, and 0.8 indicates a large effect. Therefore, ultrasound guidance may result in little to no difference in participant satisfaction.
First-attempt success	556 per 1000	778 per 1000 (717 to 845)	RR 1.40 (1.29 to 1.52)	5205 (50 RCTs)	⊕⊕⊕⊖ Moderate ^h	
Technical failure of neuraxial anaesthesia	38 per 1000	34 per 1000 (24 to 49)	RR 0.89 (0.62 to 1.29)	4581 (40 RCTs)	⊕⊕⊕⊖ Low ⁱ	
Pain during the procedure Scale from: 0 to 10, where 10 is maximum pain	The mean pain during the procedure was 2.88	MD 0.2 lower (0.52 lower to 0.13 higher)	-	1532 (13 RCTs)	⊕⊕⊕⊖ Very low ^{f,g,j}	The measurement scales were 11-point numeric scales (0 to 10, where 0 means no pain) in all 13 studies.

Any adverse events ^k	107 per 1000	76 per 1000 (52 to 111)	RR 0.71 (0.48 to 1.03)	1692 (15 RCTs)	⊕⊕⊕⊕ Very low ^{f,i}
---------------------------------	--------------	-----------------------------------	----------------------------------	-------------------	---------------------------------

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_450733005599962539.

^a There were four studies that were rated at overall high risk of bias. However, the weight of these studies in the meta-analysis was small and their impact on the overall results was probably minimal. Hence, we did not downgrade for risk of bias.

^b The I² statistic (91%) indicated considerable heterogeneity. However, overlap of confidence intervals, and similarity of point estimates were observed. We attributed importance to these observations, since when there are many studies in a meta-analysis, the statistical test has high power to detect a small amount of heterogeneity that may be clinically unimportant (<https://training.cochrane.org/handbook/current/chapter-10#section-10-10-2>). Hence, we did not downgrade for inconsistency.

^c There were six studies that were rated at overall high risk of bias. However, the weight of those studies in the meta-analysis was small, and their impact on the overall results was probably minimal. Hence, we did not downgrade for risk of bias.

^d The I² statistic (97%) indicated considerable heterogeneity. However, the overlap of confidence intervals, and similarity of point estimates were observed. We attributed importance to these observations, since when there are many studies in a meta-analysis, the statistical test has high power to detect a small amount of heterogeneity that may be clinically unimportant. Hence, we did not downgrade for inconsistency.

^e Measures of participant satisfaction varied across studies: five-point scale (five is very good), five-point scale (one is very good), Verbal Rating Scale (0 to 10, 10 is very good), four-point scale (four is very good), six-point scale (six is very good), three-point scale (three is very good), number of participants with satisfaction. Hence, we used SMD to to synthesize the results.

^f Downgraded one level due to risk of bias: most studies had high overall risk of bias.

^g Downgraded one level due to inconsistency: point estimates and confidence intervals vary considerably.

^h Downgraded one level due to publication bias. The funnel plot was asymmetrical.

ⁱ Downgraded two levels due to imprecision, i.e. the wide range from upper to lower confidence limits and the 95% CI overlaps no effect. Did not meet the optimal information size.

^j Downgraded one level due to imprecision, i.e. the wide range from upper to lower confidence limits and the 95% CI overlaps no effect.

^k Adverse events included: puncture site haemorrhage, unintentional dural puncture, back pain, headache, radicular pain, spinal haematoma, severe complication (such as death or irreversible disability), bloody tap or bloody puncture paraesthesia, postoperative nausea and vomiting.

BACKGROUND

Description of the condition

Neuraxial anaesthesia, which normally includes spinal, epidural, or combined spinal and epidural anaesthesia (CSEA), is a common technique for regional anaesthesia. It is mainly applied to anaesthesia for surgery or during labour. In neuraxial anaesthesia, the operator inserts the needle through the person's back and advances the needle to the target space, such as the subarachnoid or epidural space, to inject the drug or to place the catheter. Conventional neuraxial anaesthesia is performed using anatomical landmarks, such as Tuffier's line (the horizontal line connecting the superior iliac crests). Tuffier's line generally denotes the L4 or the L4/5 intervertebral space, or an imaginary line between the lower edges of the scapulae correlating with the T7 interspace [1]. However, the first-pass success rate of neuraxial anaesthesia with anatomical landmarks is reported to be only approximately 60% to 70% [2, 3].

Neuraxial anaesthesia is considered to be successful when the needle reaches the correct position and an adequate anaesthetic effect is obtained. Neuraxial anaesthesia can be difficult to administer successfully for several reasons. First, it is difficult to determine the point of needle insertion in people with poorly palpable landmarks and positioning challenges (e.g. obese people, pregnant women, and elderly people). Second, the operator may also miscount the target intervertebral space. It is reported that proper identification of the actual vertebral level by palpation is only about 30% [4, 5]. Third, the needle may not accurately reach the epidural or subarachnoid space because the anatomical position imagined by palpation can deviate from the actual position. Such inaccuracies can result in multiple needle attempts and longer procedures. Multiple needle passes and attempts may lead to back pain, postdural puncture headache, paralysis, and spinal haematoma, which may cause permanent neurological impairment [6]. Spinal haematoma is rare, but once it occurs, partial recovery is reported at 28% and no recovery at 25% [7]. Therefore, it is important to succeed with as few punctures as possible. Increasing successful neuraxial anaesthesia can help to reduce complications and increase patient satisfaction. Neuraxial anaesthesia by ultrasound guidance may be a useful way to increase success rates.

Description of the intervention and how it might work

Bogin first reported the use of ultrasound for lumbar puncture in 1971 [8]. Cork subsequently introduced the use of ultrasound for epidural anaesthesia in 1980 [9]. By using ultrasound, we can obtain helpful information about performing neuraxial anaesthesia, including the identity of a given intervertebral level, the depth of the epidural space, and the location of the midline and interspinous/interlaminar spaces. The main views of ultrasound techniques are the paramedian and midline views of the spine. The paramedian view is used to count upward from the sacrum and to identify the target intervertebral space. The midline transverse view is then used to show points between the spinous processes, such as the transverse process, dura, subarachnoid space, vertebrae, etc. We can measure the distance from the skin to the dura and the angle of insertion of the needle with this view [10].

Two ultrasound-guided techniques are usually used in neuraxial anaesthesia. The first is a preprocedural technique. In this

technique, the operator uses ultrasound to estimate the point of needle insertion and the depth of the target, then marks the insertion point. In this case, ultrasound is used before the needle is inserted. Markings may move to the wrong place before the needle is inserted, and the procedure is still done under the blind technique at the time of the puncture. Another method uses a newer, real-time technique [11, 12]. It is not widespread because there are technical challenges that include obtaining alignment, adequate needle visualisation, ultrasound beam, or target area [13], and operator training is required. However, this technique may increase the reliability and safety of the procedure because anaesthetists can guide the movement of the needle under ultrasound and observe the injection of the drug.

Ultrasound provides beneficial imaging that allows the operator to visualise the local anatomy and the size of important structures, and helps to determine the depth and direction of needle insertion. In addition to the operator, another anaesthetist can review the images. In children, there is some evidence that ultrasound guidance leads to improved success of neuraxial anaesthesia [14]. Ultrasound guidance may also increase success rates and prevent complications in adults, especially in those in whom neuraxial anaesthesia is deemed difficult. Other useful imaging tools for epidural anaesthesia are fluoroscopy and CT guidance [3, 15], but these require the use of radiation. Radiation can cause injuries to the skin and underlying tissues. In contrast, ultrasound is minimally invasive.

Why it is important to do this review

The efficacy of ultrasound has been established for peripheral nerve blocks [16], and its use is recognised as standard practice. On the other hand, ultrasound guidance has not been used routinely in performing neuraxial anaesthesia for surgery or labour. Previous systematic reviews and meta-analyses suggest that ultrasound guidance can reduce the risk of neuraxial anaesthesia technical failure [6, 17, 18]. However, those reviews only included the lumbar region and included diagnostic lumbar puncture, a non-anaesthetic neuraxial procedure. Another systematic review and meta-analysis was published in 2020, but it only summarised studies in obstetrics [19]. Hence, the efficacy and safety of ultrasound guidance for neuraxial anaesthesia have not been sufficiently established. In addition, some issues were not addressed in those reviews. First, it is unclear whether ultrasound improves patient-reported outcomes, such as self-reported satisfaction. More recent randomised controlled trials have reported those outcomes [20, 21], but they were published after the previous reviews. Second, efficacy for people with all body types is undetermined. For example, some studies reported no advantage of ultrasound in anatomically-easier people, such as those who are not obese or who have easily palpable spines [22, 23]. Third, to date, there is no synthesised evidence for real-time techniques. Randomised controlled trials focused on the real-time technique have recently been published, but they reported conflicting results [24, 25]. Thus, a methodologically rigorous meta-analysis is needed to evaluate people for whom ultrasound-guided neuraxial anaesthesia is beneficial, and to report outcomes of interest to people undergoing the procedure. After our protocol publication, some reviews were published [26, 27, 28]. We will discuss these reviews in [Agreements and disagreements with other studies or reviews](#).

Introducing the ultrasound guidance technique is difficult in some countries, such as low-resource countries, because the cost of purchasing an ultrasound machine is approximately USD 25,000 or more [29]. If the use of ultrasound for neuraxial anaesthesia is shown to improve outcomes, policymakers can consider funding ultrasound machines to encourage ultrasound-guided neuraxial anaesthesia. For healthcare providers, evaluation of efficacy and safety can help prioritise education and clinical resources for neuraxial anaesthesia.

OBJECTIVES

To assess the clinical efficacy and safety of ultrasound guidance compared with anatomical landmarks for neuraxial anaesthesia in adults.

METHODS

We followed the Methodological expectations for Cochrane intervention reviews (MECIR) when conducting the review [30], and PRISMA 2020 for the reporting [31]. After the publication of our protocol, some methods were modified (see [Supplementary material 9](#)).

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs), including cluster-RCTs. We included studies reported in full text, those published as abstracts only, and unpublished data. We did not include cross-over trials and quasi-RCTs (studies in which participants are allocated based on alternation, use of alternate medical records, date of birth, or other predictable methods).

Types of participants

We included all adult participants (≥ 18 years) who required neuraxial anaesthesia (spinal, epidural, or combined spinal and epidural) for surgical anaesthesia (alone or in combination with general anaesthesia) or labour. Epidural anaesthesia at both the thoracic and lumbar levels were included.

We excluded studies reporting data from children (< 18 years) because their anatomy may differ from adults, and the effect of ultrasound guidance may be different for them. We excluded studies in which neuraxial anaesthesia was used to treat chronic pain, because the target population and the procedure itself may be different from surgical anaesthesia. If we identified studies that included only a subset of eligible participants, we extracted the relevant data from that subset. We contacted the study authors to request data when stratified results were not provided. We excluded studies if more than 30% of participants were ineligible, or if data for the eligible subset were not available.

Types of interventions

We included studies comparing ultrasound guidance with anatomical landmarks when performing neuraxial anaesthesia. Ultrasound guidance included both preprocedural and real-time techniques. We included studies regardless of the level of training of operators, the number of operators (one or two), concentration of local anaesthetics used, ultrasound manufacturer, or ultrasound machine generation. We included the use of any anatomical landmarks that can be identified without tools (e.g. Tuffier's line) as

a comparison, and excluded the use of CT, fluoroscopy, or any other identification tools.

We excluded studies that evaluated computer-aided three-dimensional ultrasound because the intervention was a different approach to conventional ultrasound. We excluded studies on lumbar puncture, which is usually performed in the emergency department, because participant characteristics and the rationale for performing the procedure are expected to be different from those of neuraxial anaesthesia for surgery or labour. We also excluded any other non-anaesthetic neuraxial procedures.

Outcome measures

Critical outcomes

- Number of attempts until success or procedure termination
- Procedure time
- Participant satisfaction during procedure

The definition of 'number of attempts' varied between studies. For example, it was defined as 'any forward needle advancement following a backward movement in order to redirect the needle' in some studies (e.g. [22]), or as 'the number of insertions through the skin' in other studies (e.g. [32]). We extracted 'number of insertions through the skin' as a priority because it is more relevant for clinical decision-making.

Procedure time was reported by the included studies as preparation time, needling time, or the total of both. We only extracted outcome data from studies that reported needling time because needling time is considered to be most relevant for clinical decision-making.

Participant satisfaction was reported using a scale, such as a verbal rating scale [33]. We applied no restrictions regarding the types of scales used, and extracted both continuous and dichotomous outcome data.

Important outcomes

- First-attempt success
 - We extracted data on 'first skin insertion' as a priority.
- Technical failure of neuraxial anaesthesia, defined as any of the following situations, regardless of 'pass success':
 - Conversion to general anaesthesia
 - Discontinuation of the techniques
 - The intended anaesthetic effect was not achieved (e.g. additional analgesia required)
- Pain during procedure
 - We regarded 'pain during procedure' in the same way as participant satisfaction.
- Any adverse events, as reported by the study investigators, e.g. back pain, postdural puncture headache, radiculopathy, spinal haematoma, lasting neurological injury, and life-threatening events

Search methods for identification of studies

All search methods and search strategies are presented in [Supplementary material 1](#). After the original search on 11 October 2022, we updated the electronic searches on 28 November 2023. We did not use any date or language restrictions for trials.

Electronic searches

We aimed to identify all relevant RCTs and cluster-RCTs, regardless of language or publication status (published, unpublished, in press, or in progress) according to the *Cochrane Handbook of Systematic Reviews of Interventions* [34].

We searched the following databases for relevant trials:

- The Cochrane Central Register of Controlled Trials (CENTRAL; 2023, Issue 11), in the Cochrane Library (searched 28 November 2023)
- MEDLINE Ovid SP (1946 to 28 November 2023)
- Embase Ovid SP (1974 to 1 December 2023)
- Web of Science (1945 to 28 November 2023)

They were combined with adaptations of the highly sensitive search strategy designed by Cochrane to identify RCTs and controlled clinical trials (see [Supplementary material 1](#)).

We also conducted a search of ClinicalTrials.gov and the World Health Organization (WHO) International Clinical Trial Registry Platform (ICTPR; apps.who.int/trialsearch) for ongoing or unpublished trials, on 28 November 2023.

Searching other resources

We checked the citations of included studies and systematic reviews for additional potentially relevant studies, and examined any post-publication amendments (such as errata, retractions) to identify any problematic studies. We also scanned conference abstracts of societies related to the topic, on 3 December 2023. If necessary, we contacted trial authors for additional information. We also contacted field specialists and searched for medical ultrasound companies' websites (e.g. Canon, Fujifilm, GE Healthcare, Mindray, Mobissom, Philips, Samsung, Siemens) to enquire about relevant ongoing or unpublished studies.

Data collection and analysis

Selection of studies

We used the search strategy described above to obtain titles and abstracts of the studies. Two or more review authors (YM, KM, and DR) independently screened titles and abstracts of the results of the search to identify potentially relevant studies. We obtained the full text of these studies, if necessary, to determine which studies met eligibility criteria. We contacted the study authors if there was any missing information to determine study eligibility. When we did not receive a response, the study was classified as 'awaiting classification'. We resolved conflicts by discussion. If we were unable to reach a consensus, we consulted another review author (NW). We recorded the selection process in sufficient detail to complete a PRISMA flow diagram and characteristics of excluded studies table [31].

Data extraction and management

We developed a standardised data collection form for study characteristics and outcome data in Covidence [35], and pilot tested it on at least one included study, resolving any disagreements through discussion.

- Methods: study design, total duration of study, number of study centres and country, study setting, and date of study

- Participants: number randomised, number lost to follow-up/withdrawn, number analysed, mean age, age range, gender, body mass index (BMI), difficulty of palpating landmarks, type of surgery or procedure, inclusion criteria, and exclusion criteria
- Interventions: intervention and comparator characteristics, intervertebral level, route of approach (i.e. median or paramedian), experience of the operators, types of needle, the types of probe for ultrasound, the manufacturer or generation of ultrasound machines used, concomitant medications, and excluded medications
- Outcomes: critical and important outcomes specified and collected, and time points reported
- Note: funding for the trial, and notable conflicts of interest of trial authors

Four review authors (YM, KM, SY, and DR) independently extracted the data using standard data extraction forms. Studies reported in journals in languages other than English were translated prior to assessment. If multiple publications of a single study were identified, we grouped the reports together and extracted the most likely correct data from multiple reports for analysis. We resolved conflicts about data extraction by discussion. If we were unable to reach a consensus, we consulted another review author (NW). If additional information was needed, one review author (YM) contacted the corresponding author of the relevant studies. After we had completed data extraction, one review author (YM) transferred the data to RevMan [36], and another review author (IN) checked the data.

Risk of bias assessment in included studies

Two of the review authors (YM, KM, SY, or DR) independently assessed the risk of bias for each included study using the Cochrane RoB 2 tool for randomised trials [37]. We assessed the risk of bias for the effect of assignment to the interventions at baseline (i.e. the intention-to-treat effect) for the seven critical and important outcomes (see [Outcome measures](#)). We used the RoB 2 Excel tool to manage the assessment of bias (available from www.riskofbias.info/welcome/rob-2-0-tool/current-version-of-rob-2). We resolved conflicts by discussion. If we were unable to reach a consensus, we consulted another review author (NW).

We assessed the following RoB 2 domains, using a series of signalling questions:

- Bias arising from the randomisation process;
- Bias due to deviations from intended interventions;
- Bias due to missing outcome data;
- Bias in measurement of the outcome; and
- Bias in selection of the reported result.

We assessed all domains for each outcome across different studies, and judged each domain according to these risk-of-bias ratings:

- Low risk of bias;
- Some concerns; or
- High risk of bias.

We summarised the overall risk of bias for each outcome as described in the *Cochrane Handbook* [37]. If we identify and include eligible cluster-RCTs in future updates, we will use the RoB 2 tool

for cluster-RCTs, and assess an additional domain specific to this study design (bias arising from the timing of identification and recruitment of participants) [38].

We reached an overall risk-of-bias judgement for an outcome as follows:

- Low risk of bias: all the above domains were judged to be at low risk of bias.
- Some concerns: one or more of the above domains were judged to raise some concerns, but none were judged to be at high risk of bias.
- High risk of bias: one or more of the above domains were at high risk of bias, or multiple domains raised some concerns in a way that substantially lowered confidence in the result.

Measures of treatment effect

We analysed dichotomous data as risk ratios (RR) with 95% confidence intervals (CIs); continuous data as mean difference (MD) for continuous data using the same scale, and standardised mean difference (SMD) for continuous data using different scales, with 95% CIs. We used SMD for the outcome, participant satisfaction. We interpreted the results of SMD as follows: 0.2 indicated a small effect, 0.5 indicated a medium effect, and 0.8 indicated a large effect [39]. If a trial reported continuous data as median and interquartile range (IQR), we converted the median to mean and estimated the standard deviation (SD) as $IQR/1.35$ [40]. When the data were skewed, we used a formula for imputing a missing mean value based on the lower quartile, median, and upper quartile summary statistics proposed by Wan and colleagues, which has been reported to work well in analysing skewed outcomes [41].

Unit of analysis issues

The unit of analysis was the individual participant. If we include cluster-RCTs in future updates, we will consider the cluster as the unit of randomisation, and adjust reported standard errors using the trial's intracluster correlation coefficient (ICC), using the method described in the *Cochrane Handbook* [40]. If we cannot obtain the ICC, we will use an approximate ICC from a similar trial. For studies with multiple treatment groups, we included all intervention groups relevant to the review. For RCTs that had two intervention groups and one comparison group, we divided the number of participants in the comparison group by the number of intervention groups to avoid double counting the participants when they were included in the same analysis.

Dealing with missing data

We contacted the original investigators or study sponsors to verify key study characteristics and to obtain missing numerical outcome data, where possible (e.g. when a study was identified as an abstract only). Where this was not possible, and the missing data were thought to introduce serious bias, we explored the impact of including such studies in the overall assessment of results by a sensitivity analysis.

Reporting bias assessment

We created and examined funnel plots to explore possible small-study biases. We also evaluated reporting bias by checking the protocol of the study, if we could identify one from searching trial registries. We performed the Egger test to test for funnel plot asymmetry [42].

Synthesis methods

We reviewed the data from all eligible studies and synthesised and analysed the data using RevMan [36]. For dichotomous outcomes, we used the number of participants in each trial arm and the number of events. For continuous outcomes, we used means and SDs for each trial arm. We reported data narratively if it was inappropriate to combine them in a meta-analysis. We undertook a meta-analysis only where this was meaningful, i.e. if the treatments, participants, and the underlying clinical question were similar enough for pooling. We used a random-effects model to pool data because we expected the definitions of participants and operators to vary to some extent between studies.

We visually inspected forest plots to consider the direction and magnitude of effects and the degree of overlap between confidence intervals. We used the I^2 statistic to measure heterogeneity amongst the trials in each analysis, but acknowledge that there is substantial uncertainty in the value of I^2 when there are few studies. Therefore, we also considered the P value from the χ^2 test, with a significance level of the P value set to 0.10 [43]. If we identified substantial heterogeneity, we investigated and reported potential reasons for this by prespecified subgroup analysis. We interpreted the heterogeneity as follows.

- 0 to 40%: might not be important
- 30 to 60%: may represent moderate heterogeneity
- 50% to 90%: may represent substantial heterogeneity
- 75% to 100%: considerable heterogeneity

When I^2 lay in an area of overlap between two categories (e.g. between 50% and 60%), we considered differences in participants and interventions amongst the trials that contributed data to the analysis [43].

Investigation of heterogeneity and subgroup analysis

We planned the following subgroup analysis.

- Difficulty in palpating landmarks, such as in people with obesity; people with scoliosis or previous spinal surgery, and elderly people: difficult versus not difficult
- Experience of operators (e.g. Residents, Fellows): experienced versus inexperienced
- Types of ultrasound guidance: pre-procedural versus real-time techniques
- Types of neuraxial anaesthesia: spinal versus lumbar epidural versus thoracic epidural
- Definitions of 'number of attempts' or 'first-attempt success'

Because we planned to explore possible causes of substantial heterogeneity with subgroup analysis, we used all outcomes in subgroup analyses. We used the formal test for subgroup differences in RevMan [36], and based our interpretation on this.

Equity-related assessment

We did not examine any equity-related analysis in this review. We described this issue in [Limitations of the evidence included in the review](#).

Sensitivity analysis

We planned to carry out the following sensitivity analyses, to test whether key methodological factors or decisions affected the main result.

- Including only overall low risk of bias results and excluding results with some concerns or overall high risk of bias
- Excluding studies with incomplete or imputed data, such as mean or SD

Certainty of the evidence assessment

We constructed a summary of findings table using GRADEpro GDT software, and by following Cochrane guidance [44]. The summary of findings table included the relative and absolute effects and the certainty of the evidence for all seven critical and important outcomes (see [Outcome measures](#)). Two review authors (YM, KM), working independently and in duplicate, applied GRADE criteria to evaluate the certainty of the evidence for each outcome [45]. We downgraded the level of certainty based on risk of bias/study limitations (using the overall risk of bias as determined by the RoB 2 tool; see [Risk of bias assessment in included studies](#)), imprecision, inconsistency, indirectness, and publication bias. We resolved conflicts by discussion. If we were unable to reach a consensus, we consulted another review author (NW). In the case of credible subgroup effects based on characteristics of the population, we presented results stratified across subgroups in the summary of findings table.

Consumer involvement

Consumers were not involved in this review. Four review authors (YM, KM, DR, IN) have expertise in anaesthesia and considered outcomes important to healthcare providers and patients.

RESULTS

Description of studies

Results of the search

The first search in October 2022 identified 11,813 records; the updated search in November 2023 identified 1819 records from the electronic databases. Through screening the references of relevant studies and systematic reviews, we identified another three potentially eligible records, in addition to the existing 13,632 (Bal 2019 [46]; Balaban 2017 [47]; Vernon 2020 [48]). After the removal of duplicates, titles and abstracts of 8571 records were screened to remove clearly irrelevant studies. Subsequently, we retrieved full-text reports of 207 records for eligibility assessment. We included 65 studies (98 records) in our review. Since one study only reported the incidence of complications (not the number of participants) (Wilkes 2017 [49]), our meta-analysis included data from 64 studies. We also identified six ongoing studies, and 24 studies are awaiting classification. [Figure 1](#) shows our study selection process.

Figure 1. PRISMA study selection flow diagram * Characteristics of 15 excluded studies are listed in [Supplementary material 3](#).

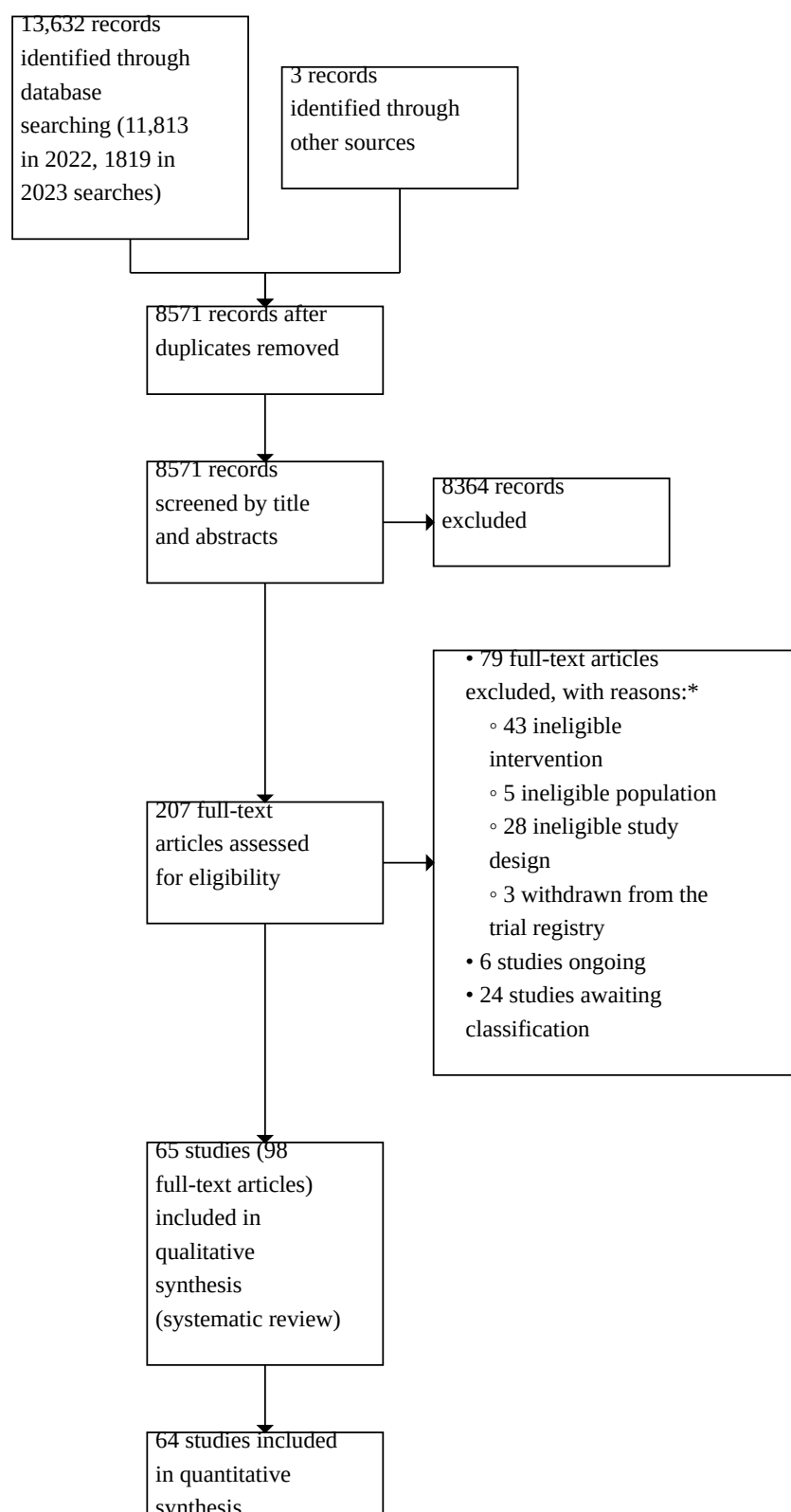


Figure 1. (Continued)

in quantitative
synthesis
(meta-analysis)

Included studies

We included 65 studies, involving 6823 participants (Abdelhamid 2013 [50]; Ansari 2014 [33]; Arzola 2015 [22]; Arzola 2022 [51]; Auyong 2017 [52]; Bal 2019; Balaban 2017; Bhardwaj 2022 [53]; Chin 2011 [54]; Chin 2018 [55]; Creaney 2016 [56]; Dhanger 2018 [57]; Ekinici 2017 [58]; Elsharkawy 2017 [25]; Galal 2019 [59]; Gayathri 2021 [60]; Geng 2016 [61]; Gorle 2023 [62]; Grau 2001a [63]; Grau 2001b [64]; Grau 2002 [65]; Grau 2004 [66]; Hasanin 2017 [67]; Isiker 2023 [68]; Jagadish 2021 [69]; Jain 2019 [70]; Jayanth 2023 [71]; Jiang 2021 [72]; Kampitak 2018 [73]; Kawaguchi 2011 [74]; Khan 2022 [75]; Korda 2020 [76]; Kumar 2020 [77]; Li 2019 [20]; Lim 2012 [78]; Narkhede 2019 [79]; Nassar 2015 [80]; NCT02207972 [81]; NCT02553746 [82]; Pakpirom 2022 [83]; Park 2019 [21]; Park 2020 [84]; Perna 2017 [85]; Pramodrao 2022 [86]; Qu 2020 [87]; Rahimzadeh 2022 [88]; Ray 2022 [89]; Rizk 2019 [90]; Sahin 2014 [32]; Singh 2023 [91]; Srinivasan 2015 [92]; Srinivasan 2018 [93]; Tao 2021 [94]; Tawfik 2017 [95]; Tawfik 2023 [96]; Tubinis 2019 [97]; Turkstra 2017 [23]; Urfalioglu 2017 [98]; Uyel 2021 [99]; Vallejo 2010 [100]; Vernon 2020; Wang 2012 [101]; Wang 2014 [102]; Wilkes 2017; Wu 2021 [103]).

Table 1 illustrates an overview of included studies and syntheses. Comprehensive study characteristics are available in [Supplementary material 2](#).

Study designs and settings

All studies were parallel randomised controlled trials (RCTs), each conducted at a single centre. Three studies were three-arm trials (two types of intervention and one control) (Bhardwaj 2022; Grau 2004; Rizk 2019), and one study was a four-arm trial (two types of intervention and two types of control) (Sahin 2014). Studies were conducted in Africa, Asia, Europe, the Middle East, North America, and Oceania, and published between 2001 and 2023. All studies were conducted in tertiary or university hospitals, and were conducted at a single institution, except for one study, which was conducted at two hospitals (Turkstra 2017). We did not identify any studies sponsored by industry. We extracted study details and results for two studies from the trial registry, since they were not yet published (NCT02207972; NCT02553746). One study identified by the database search was available in full text as the author's Masters' thesis (Korda 2020). However, we were unable to find it published in any journals.

Study participants

The mean or median age ranged from 23 to 83 years. Body mass index (BMI) ranged from 32 kg/m² to 46 kg/m². Thirty-two studies evaluated obstetric populations with labour epidural or caesarean section (3149 participants), while the remaining studies evaluated participants who received surgery under neuraxial anaesthesia. Most of them underwent lower abdominal or lower limb surgery.

Four studies evaluated upper abdominal or thoracic surgery using thoracic epidural anaesthesia (Arzola 2022; Auyong 2017; Hasanin 2017; Pakpirom 2022). Thirty-three studies evaluated spinal anaesthesia (3309 participants), 15 studies evaluated lumbar epidural anaesthesia (1530 participants), 12 studies evaluated combined spinal and epidural anaesthesia (CSEA; 1552 participants), and four studies evaluated thoracic epidural anaesthesia (261 participants). One study (146 participants) included multiple types of neuraxial anaesthesia (NCT02553746). For the purpose of the subgroup analysis by types of neuraxial anaesthesia, we classified studies that evaluated CSEA as lumbar epidural anaesthesia. Details about the sex, age, BMI, inclusion and exclusion criteria for each study are shown in [Supplementary material 2](#).

Interventions and comparisons

Four studies evaluated real-time ultrasound as an intervention arm (Elsharkawy 2017; Galal 2019; Jiang 2021; Pakpirom 2022). Two studies were three-arm trials, which evaluated pre-procedural ultrasound and real-time ultrasound (Bhardwaj 2022; Grau 2004). Ultrasound machines used in the included studies were manufactured by various companies. Most studies used convex or curvilinear probes in the ultrasound groups. Details about the needle approach (median or paramedian), operator, types of needles, ultrasound machine, and probes for each study are shown in [Supplementary material 2](#).

Outcome measures

Studies that contributed results to our data synthesis are summarised in Table 1. For continuous outcomes, many studies reported median and interquartile range (IQR), instead of mean and standard deviation (SD). We converted these results using the methods detailed in [Dealing with missing data](#). For participant satisfaction, various measurement scales, some indicating the opposite direction, were used in each study. Some studies reported the number of participants who felt satisfied. We synthesised participant satisfaction using the standardised mean difference (SMD).

Excluded studies

We excluded 79 full-text reports. The reasons for excluding 15 of these reports are documented in [Supplementary material 3](#). Six studies were excluded for ineligible study design; five studies for ineligible intervention, which was computer-aided three-dimensional ultrasound; two were excluded because computed tomography (CT) or X-ray was used in the control group; one study did not address the outcomes of interest; and in one study, the participants did not receive anaesthesia.

Details of studies classified as awaiting classification, and ongoing studies are shown in [Supplementary material 4](#) and [Supplementary material 5](#).

Risk of bias in included studies

We included the risk of bias assessments and support for judgements across all RoB 2 domains for each outcome in [Supplementary material 6](#). The risk of bias tables for each analysis are shown next to the forest plots in [Supplementary material 7](#). Our detailed risk-of-bias assessments are available upon request.

We judged 46 studies at low risk of bias in the randomisation process. Some studies raised some concerns, mainly because methods of randomisation or allocation concealment were not described. Amongst those studies, group differences at baseline did not suggest a problem with randomisation. Regarding the 'selection of the reported results' domain, nearly half of the studies had no publicly available trial registration or study protocol. Those studies raised some concerns in the domain across all outcomes. Our risk-of-bias assessments at outcome level are described below.

Critical outcomes

Number of attempts until success or procedure termination

Amongst the 57 studies that contributed data on this outcome, four studies had an overall high risk of bias (Elsharkawy 2017; Geng 2016; Korda 2020; Vernon 2020). The domain that contributed to the high risk of bias was missing outcome data (domain 3 of the RoB 2 tool), mainly because participants with difficulty or failure in neuraxial anaesthesia were excluded, which led to missing data. Other studies that raised some concerns were because of bias arising from the randomisation process (domain 1) or bias in selection of the reported results (domain 5), as mentioned above.

Procedure time

Of the 43 studies included in the analysis, six studies had an overall high risk of bias (Elsharkawy 2017; Geng 2016; Korda 2020; Rahimzadeh 2022; Tao 2021; Vernon 2020). Domains that contributed to a high risk of bias were deviations from intended interventions (domain 2) or missing outcome data (domain 3), mainly because participants with difficulty or failure in neuraxial anaesthesia did not receive the intervention or were excluded from analysis. Other studies that raised some concerns were because of bias arising from the randomisation process (domain 1) or bias in selection of the reported results (domain 5).

Participant satisfaction during the procedure

Twenty of the 25 studies that reported this outcome were judged to be at an overall high risk of bias. The domain that contributed to a high risk of bias was measurement of the outcome (domain 4), mainly because outcome assessors were probably aware of the intervention. Since this is a participant-reported outcome, we consider that measurement/assessment of the outcome had been influenced by prior knowledge of the intervention received. Only five studies were judged to have low risk bias in measurement of the outcome (domain 4), because participants were probably unaware of their assigned intervention, based on the performance of sham ultrasound in the control group (Jiang 2021; Kampitak 2018; Pakpirom 2022; Tawfik 2017; Tawfik 2023).

Important outcomes

First-attempt success

Of the 50 studies that reported first-attempt success, six studies were judged to be at an overall high risk of bias (Geng 2016; Isiker 2023; Korda 2020; NCT02553746; Tao 2021; Vernon 2020). Domains that contributed to a high risk of bias were deviations from intended interventions (domain 2) or missing outcome data (domain 3), mainly because participants with difficulty or failure in neuraxial anaesthesia did not receive the intervention, or were excluded from analysis. Other studies that raised some concerns were because of bias arising from the randomisation process (domain 1), or bias in selection of the reported result (domain 5).

Technical failure of neuraxial anaesthesia

Amongst the 40 studies included in the meta-analysis, four studies were judged to be at an overall high risk of bias (Elsharkawy 2017; NCT02553746; Tao 2021; Vernon 2020). Domains that contributed to a high risk of bias were deviations from intended interventions (domain 2), or missing outcome data (domain 3), mainly because participants with difficulty or failure in neuraxial anaesthesia did not receive the intervention, or were excluded from analysis. Other studies that raised some concerns were because of bias arising from the randomisation process (domain 1), or bias in selection of the reported result (domain 5).

Pain during the procedure

Eleven of the 13 studies included in the analysis of pain during procedure were judged to be at an overall high risk of bias. The domain that contributed to a high risk of bias was bias in the measurement of the outcome (domain 4), is similar to our observation of the outcome 'participant satisfaction'. Only two studies were judged to be at low risk bias in this domain, since the study investigators performed sham ultrasound in the control group (Chin 2018; Kampitak 2018).

Any adverse events

Nine of the 15 studies that reported adverse events were judged to be at an overall high risk of bias. The domain that contributed to a high risk of bias was bias in measurement of the outcome (domain 4), because this outcome included a participant-reported outcome, such as a headache. Other studies that raised some concerns were because of bias in selection of the reported result (domain 5). Only three studies were judged to be at an overall low risk of bias (Chin 2018; Khan 2022; Pakpirom 2022).

Synthesis of results

Absolute and relative effects for all critical and important outcomes are summarised with their GRADE ratings, signifying the level of certainty in the effect estimates in [Summary of findings 1](#).

Critical outcomes

Number of attempts until success or procedure termination

Of the 65 included studies, 57 studies reported the number of attempts until success. The definition of this outcome varied between studies. Thirty-six studies reported both the number of insertions through the skin and the number of forward needle advancements following a backward movement. Nine studies reported only the latter (Arzola 2022; Creaney 2016; Grau 2001b;

Perna 2017; Tubinis 2019; Turkstra 2017; Vallejo 2010; Vernon 2020; Wang 2012).

Meta-analysis of the data from 6192 participants showed that ultrasound guidance compared with landmarks reduced the

number of attempts until success or procedure termination (mean difference (MD) -0.41 attempts, 95% confidence interval (CI) -0.51 to -0.31; $I^2 = 91\%$; 57 studies, 6192 participants; high-certainty evidence; Analysis 1.1, [Figure 2](#)).

Figure 2.

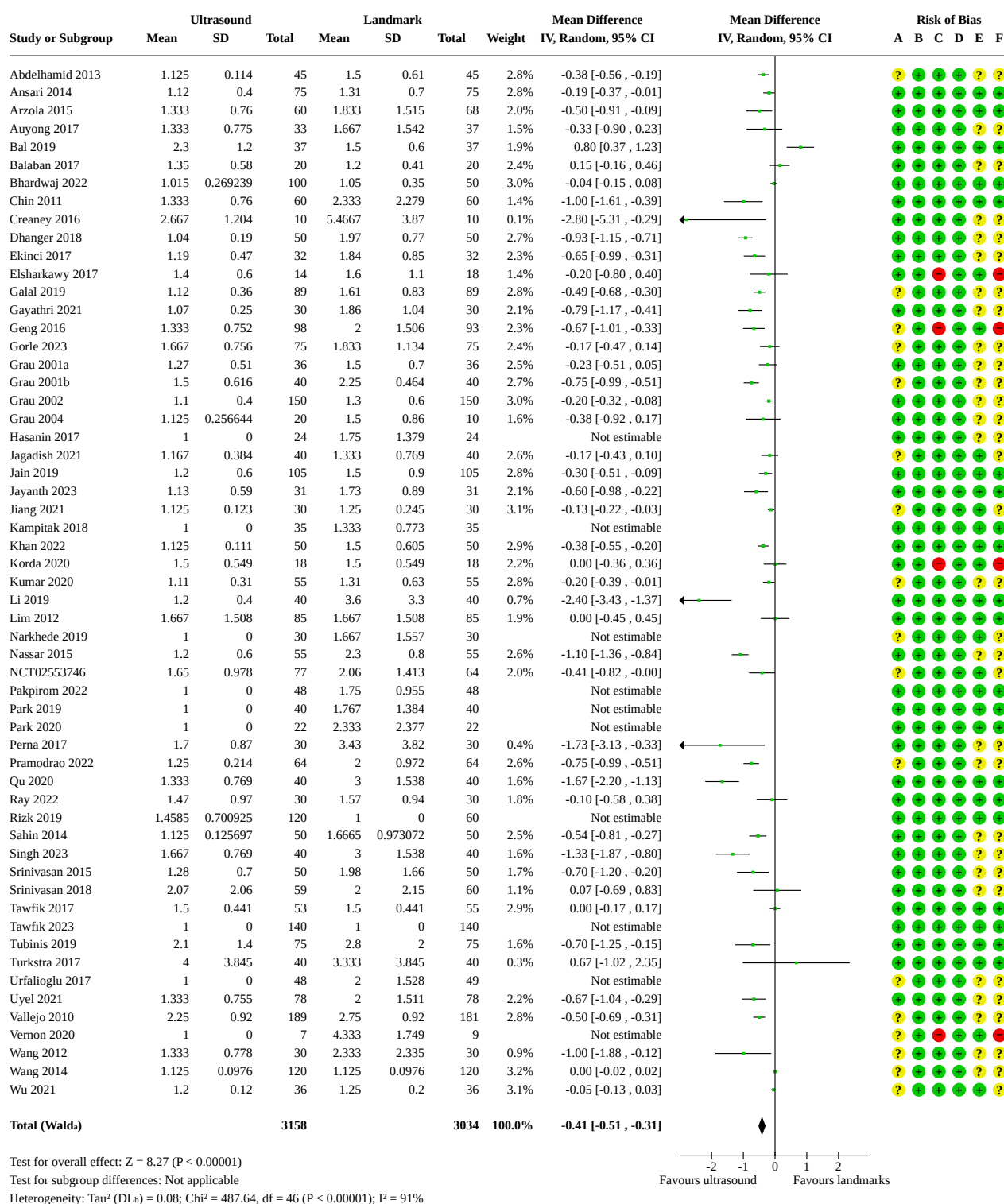


Figure 2. (Continued)
Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

We included studies that had sufficient information regarding covariates in the following prespecified subgroup analyses (see [Supplementary material 7](#)):

- Difficult versus not difficult to palpate landmarks (test for subgroup differences: $\text{Chi}^2 = 10.55$, $\text{df} = 1$, $P = 0.001$, $I^2 = 90.5\%$; Analysis 2.1);
- Experienced versus inexperienced operators (test for subgroup differences: $\text{Chi}^2 = 4.17$, $\text{df} = 1$, $P = 0.04$, $I^2 = 76.0\%$; Analysis 2.2);
- Types of ultrasound guidance: pre-procedural versus real-time techniques (test for subgroup differences: $\text{Chi}^2 = 3.79$, $\text{df} = 1$, $P = 0.05$, $I^2 = 73.6\%$; Analysis 2.3);
- Types of neuraxial anaesthesia: spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 0.20$, $\text{df} = 2$, $P = 0.90$, $I^2 = 0\%$; Analysis 2.4);
- Outcome definition: used 'number of insertions through the skin' versus 'number of forward needle advancement' (test for subgroup differences: $\text{Chi}^2 = 4.81$, $\text{df} = 1$, $P = 0.03$, $I^2 = 79.2\%$; Analysis 2.5).

There may be evidence of a difference in effect in three subgroups.

- Difficulty of palpating landmarks: the estimated MD was -0.72 attempts (95% CI -0.98 to -0.45; 21 studies, 2138 participants) for ultrasound versus landmarks in participants with difficulty; the estimated MD was -0.24 attempts (95% CI -0.35 to -0.13; 20 studies, 2034 participants) for ultrasound versus landmarks in participants without difficulty. See Analysis 2.1.
- Experience of operators: the estimated MD was -0.47 attempts (95% CI -0.60 to -0.34; 34 studies, 3230 participants) for ultrasound versus landmarks in experienced operators; the

estimated MD was -0.21 attempts (95% CI -0.42 to -0.00; 14 studies, 1799 participants) for ultrasound versus landmarks in inexperienced operators. See Analysis 2.2.

- Definition of outcome: the estimated MD was -0.65 attempts (95% CI -0.87 to -0.43; 9 studies, 964 participants) for ultrasound versus landmarks when the definition was the number of skin punctures; the estimated MD was -0.41 attempts (95% CI -0.51 to -0.31; 48 studies, 6192 participants) for ultrasound versus landmarks when the definition was not number of skin punctures. See Analysis 2.5.

The results of the planned sensitivity analyses showed minimal impact on the synthesised result (see [Supplementary material 7](#)).

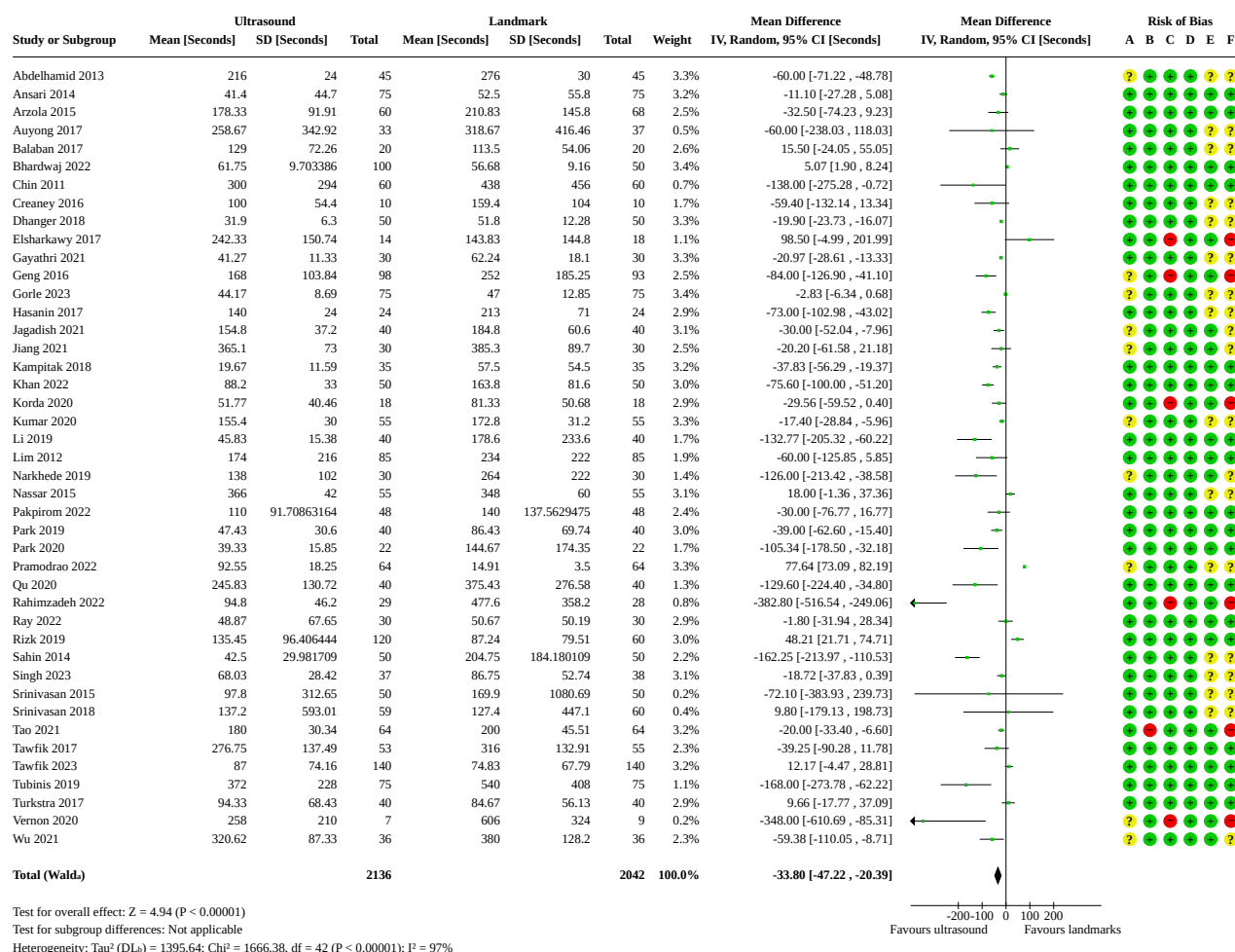
- Including only studies that were judged to be at overall low risk of bias: MD -0.18 attempts, 95% CI -0.25 to -0.12; $I^2 = 87\%$; 21 studies, 2472 participants; Analysis 2.6
- Including only studies that reported both the mean and standard deviation: MD -0.36 attempts, 95% CI -0.51 to -0.20; $I^2 = 89\%$; 21 studies, 2156 participants; Analysis 2.7

Procedure time (needling time)

Of the 65 included studies, 43 studies reported procedure/needling time. Twenty-two studies reported time in minutes. We converted the results from those studies into seconds by multiplying them by 60 before data synthesis.

Meta-analysis of the data from 4178 participants showed that ultrasound guidance compared with landmarks reduced procedure time (MD -33.8 seconds, 95% CI -47.22 to -20.39; $I^2 = 97\%$; 43 studies, 4178 participants; high-certainty evidence; Analysis 1.2, [Figure 3](#)).

Figure 3.



There were sufficient data to investigate the prespecified subgroup analyses. See [Supplementary material 7](#).

- Difficult versus not difficult (test for subgroup differences: $\chi^2 = 1.92$, $df = 1$, $P = 0.17$, $I^2 = 47.9\%$; Analysis 3.1)
- Experienced versus inexperienced (test for subgroup differences: $\chi^2 = 0.51$, $df = 1$, $P = 0.48$, $I^2 = 0\%$; Analysis 3.2)
- Pre-procedural versus real-time techniques (test for subgroup differences: $\chi^2 = 4.41$, $df = 1$, $P = 0.04$, $I^2 = 77.3\%$; Analysis 3.3)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\chi^2 = 3.50$, $df = 2$, $P = 0.17$, $I^2 = 42.8\%$; Analysis 3.4)

Results suggested a difference in effect by types of ultrasound guidance. The estimated MD was -35.98 seconds (95% CI -49.71 to -22.25; 40 studies, 3940 participants) for ultrasound versus

landmarks in pre-procedural technique; the estimated MD was 1.83 seconds (95% CI -30.69 to 34.35; 4 studies, 288 participants) for ultrasound versus landmarks in real-time technique. See Analysis 3.3.

The results of the planned sensitivity analyses showed minimal impact on the synthesised result (see [Supplementary material 7](#)).

- Including only studies that were judged to be at low risk of bias (MD -30.76 seconds, 95% CI -47.90 to -13.62; $I^2 = 88\%$; 18 studies, 2126 participants; Analysis 3.5)
- Including only studies that reported both the mean and standard deviation (MD -29.21 seconds, 95% CI -50.26 to -8.17; $I^2 = 99\%$; 21 studies, 2034 participants; Analysis 3.6)

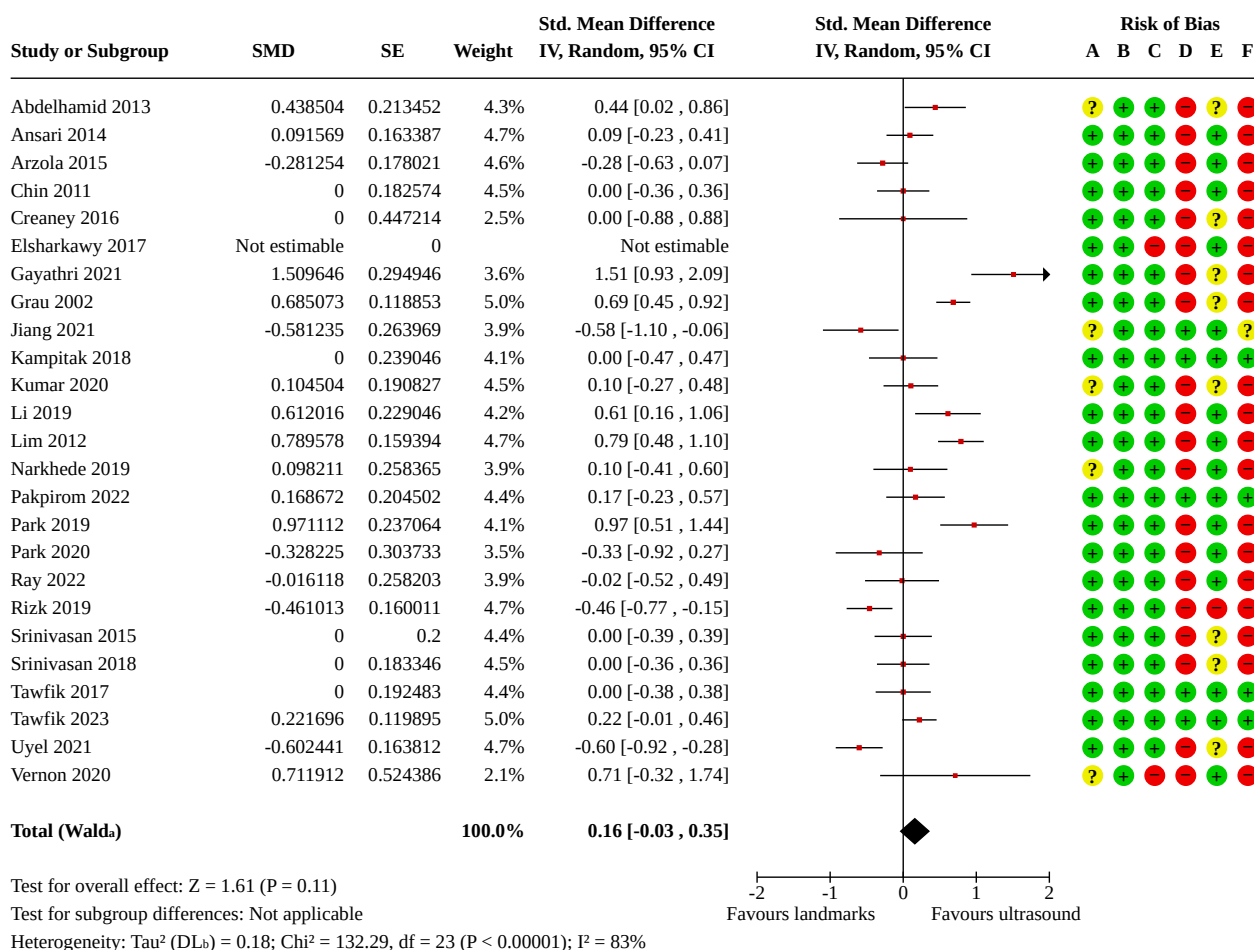
Participant satisfaction during the procedure

Of the 65 included studies, 25 studies reported participant satisfaction. The definition of this outcome varied between studies. The measurement scales also varied between studies. Some studies gave high scores for high satisfaction, others gave high scores for low satisfaction. A few studies reported only the number

of participants who reported satisfaction. We pooled the results using SMD.

Meta-analysis of the data showed that ultrasound guidance may result in little to no difference in participant satisfaction compared with anatomical landmarks (SMD 0.16, 95% CI -0.03 to 0.35; $I^2 = 83\%$; 25 studies, 2689 participants; low-certainty evidence; Analysis 1.3, Figure 4). We downgraded the certainty of evidence due to concerns about risk of bias and inconsistency.

Figure 4.



Footnotes

^aCI calculated by Wald-type method.

^b Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

There were sufficient data to investigate the prespecified subgroup analyses. See [Supplementary material 7](#). There was no evidence of a difference in effect in the investigated subgroups.

- Difficult versus not difficult (test for subgroup differences: $\text{Chi}^2 = 1.94$, $\text{df} = 1$, $P = 0.16$, $I^2 = 48.4\%$; Analysis 4.1)

- Experienced versus inexperienced (test for subgroup differences: $\text{Chi}^2 = 0.42$, $\text{df} = 1$, $P = 0.52$, $I^2 = 0\%$; Analysis 4.2)
- Pre-procedural versus real-time techniques (test for subgroup differences: $\text{Chi}^2 = 0.94$, $\text{df} = 1$, $P = 0.33$, $I^2 = 0\%$; Analysis 4.3)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 0.13$, $\text{df} = 2$, $P = 0.94$, $I^2 = 0\%$; Analysis 4.4)

The results of the planned sensitivity analyses showed that ultrasound guidance compared with landmarks increased participant satisfaction only when studies that reported both the mean and standard deviation were included, although this was an exploratory analysis (see [Supplementary material 7](#)).

- Including only studies that were judged to be at an overall low risk of bias (SMD 0.14, 95% CI -0.03 to 0.31; $I^2 = 0\%$; 4 studies; Analysis 4.5)
- Including only studies that reported both the mean and standard deviation (SMD 0.41, 95% CI 0.11 to 0.71; $I^2 = 75\%$; 9 studies; Analysis 4.6)

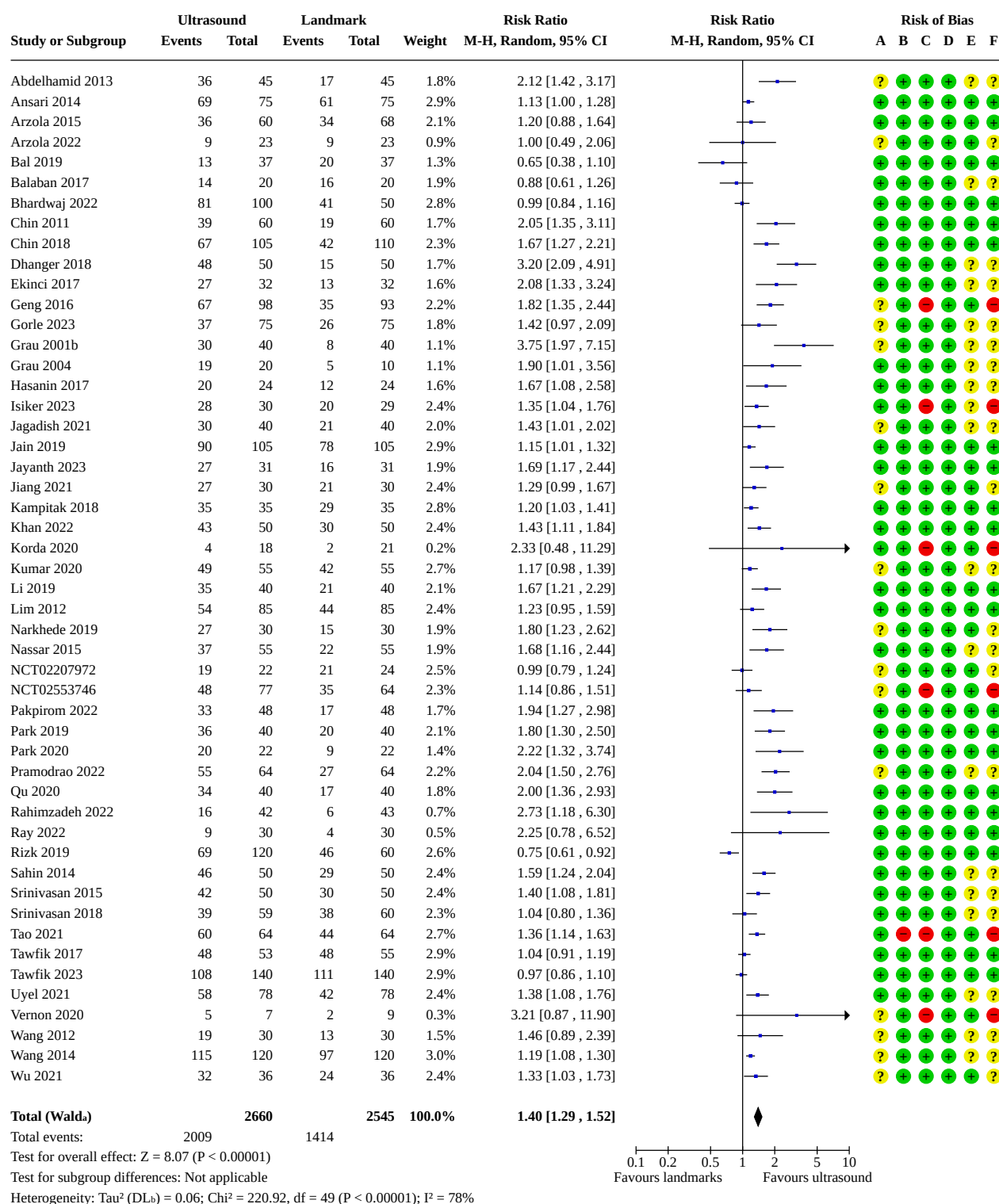
Important outcomes

First-attempt success

Of the 65 included studies, 50 studies reported first-attempt success. The definition of this outcome varied between studies. Twenty-four studies reported both the first skin puncture and the first needle advancement. Fifteen studies reported only the latter (Arzola 2015; Arzola 2022; Chin 2018; Grau 2001b; Korda 2020; Narkhede 2019; Nassar 2015; NCT02553746; Pakpirom 2022; Rahimzadeh 2022; Ray 2022; Vernon 2020; Wang 2012; Wang 2014; Wu 2021).

Meta-analysis of the data from 5205 participants showed that ultrasound guidance compared with landmarks likely increased the first-attempt success (risk ratio (RR) 1.40, 95% CI 1.29 to 1.52; $I^2 = 78\%$; 50 studies, 5205 participants; moderate-certainty evidence; Analysis 1.4, [Figure 5](#)). We downgraded the certainty of evidence for this outcome due to publication bias ([Figure 6](#)).

Figure 5.



Footnotes

^aCI calculated by Wald-type method.^bTau² calculated by DerSimonian and Laird method.

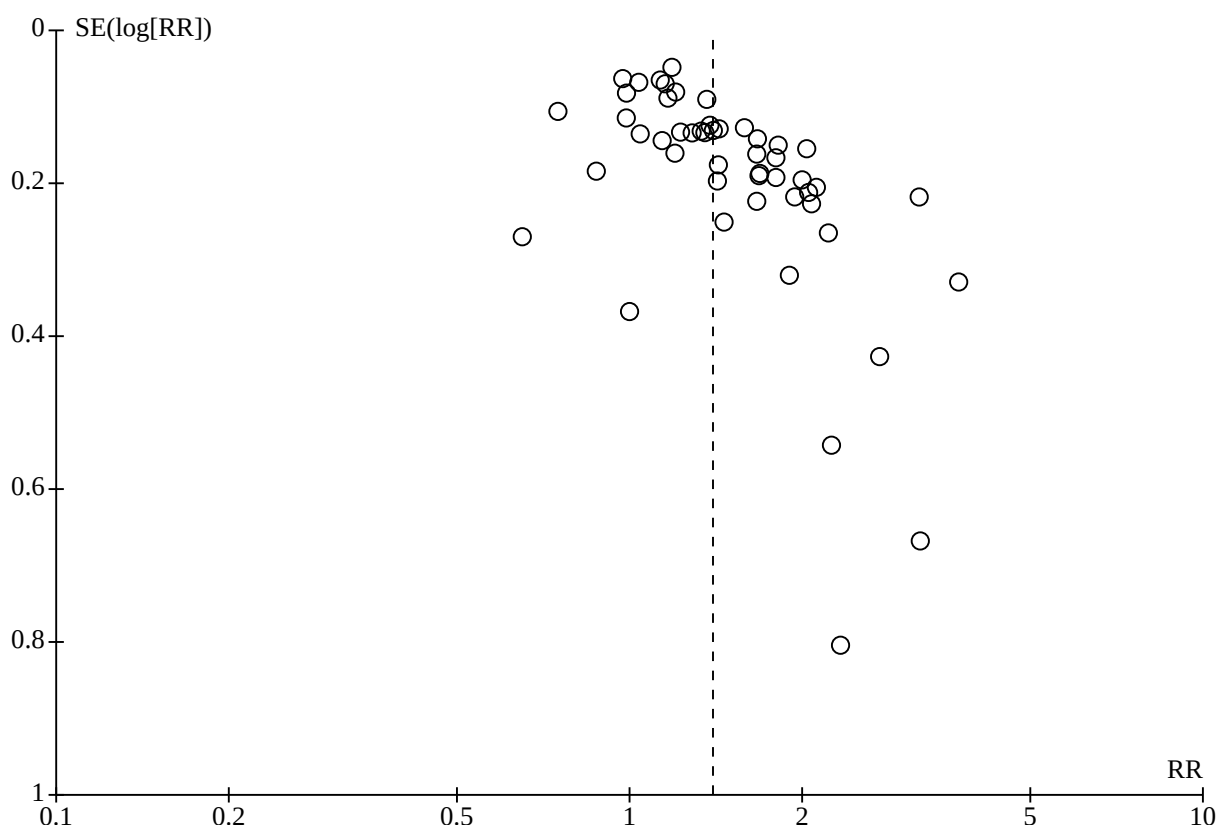
Figure 5. (Continued)

τ^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Figure 6. Funnel plot of Analysis 1.4.



There were sufficient data to investigate the prespecified subgroup analyses. There was no evidence of a difference in effect in the investigated subgroups.

- Difficult versus not difficult (test for subgroup differences: $\text{Chi}^2 = 0.54$, $\text{df} = 1$, $P = 0.46$, $I^2 = 0\%$)
- Experienced versus inexperienced (test for subgroup differences: $\text{Chi}^2 = 2.00$, $\text{df} = 1$, $P = 0.16$, $I^2 = 49.9\%$)
- Pre-procedural versus real-time techniques (test for subgroup differences: $\text{Chi}^2 = 0.00$, $\text{df} = 1$, $P = 0.98$, $I^2 = 0\%$)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 0.95$, $\text{df} = 2$, $P = 0.62$, $I^2 = 0\%$)
- Outcome definition: first skin puncture versus first needle advancement (test for subgroup differences: $\text{Chi}^2 = 1.76$, $\text{df} = 2$, $P = 0.18$, $I^2 = 43.2\%$)

The results of the planned sensitivity analyses showed minimal impact on the synthesised result. We performed only the following sensitivity analysis because we did not impute or transform the data for this outcome. See [Supplementary material 7](#).

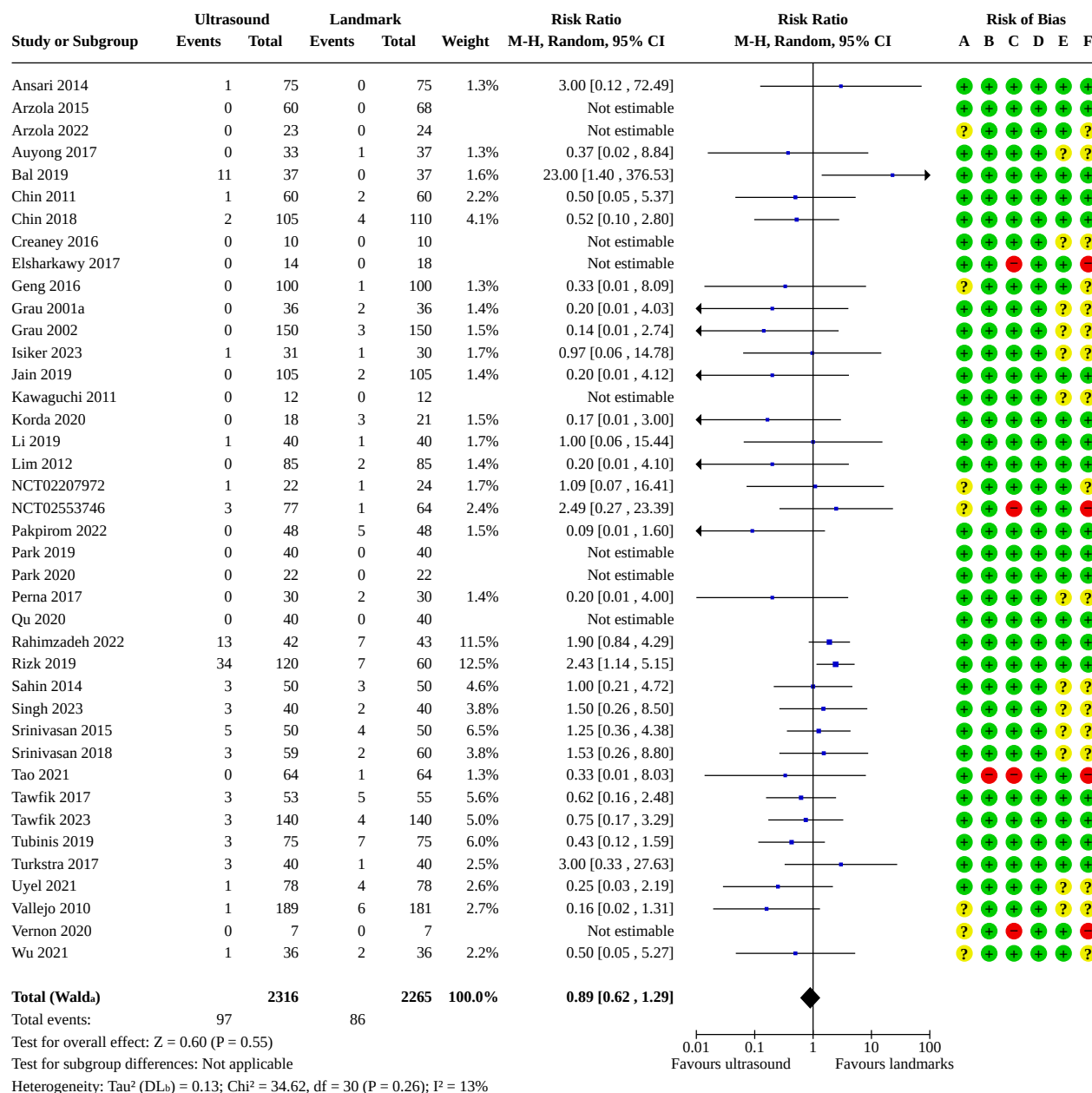
- Including only studies that were judged to be at overall low risk of bias (RR 1.34, 95% CI 1.07 to 1.67; $I^2 = 87\%$; 6 studies, 849 participants)

Technical failure of neuraxial anaesthesia

Of the 65 included studies, 40 studies reported technical failure. Meta-analysis of the data from 4581 participants showed that ultrasound guidance compared with landmarks may result in little to no difference in the risk of technical failure (RR 0.89, 95% CI 0.62 to 1.29; $I^2 = 13\%$; 40 studies, 4581 participants; low-certainty

evidence; Analysis 1.5, Figure 7). We downgraded the certainty of evidence for this outcome due to imprecision.

Figure 7.



Footnotes

^aCI calculated by Wald-type method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

There were sufficient data to investigate three of the four expected effect modifiers with subgroup analyses. We could not examine the impact of experienced versus inexperienced operators because all studies for which information on covariates was available were performed by experienced operators.

- Difficult versus not difficult (test for subgroup differences: $\text{Chi}^2 = 0.55$, $\text{df} = 1$, $P = 0.46$, $I^2 = 0\%$)
- Pre-procedural versus real-time techniques (test for subgroup differences: $\text{Chi}^2 = 2.55$, $\text{df} = 1$, $P = 0.11$, $I^2 = 60.7\%$)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 15.76$, $\text{df} = 2$, $P = 0.0004$, $I^2 = 87.3\%$)

Results suggested heterogeneity between the subgroups for the types of neuraxial anaesthesia. For ultrasound versus landmark, the estimated RR was 1.50 (95% CI 1.02 to 2.20; 20 studies, 2050 participants) in spinal anaesthesia; the estimated RR was 0.39 (95% CI 0.21 to 0.73; 16 studies, 2177 participants) in lumbar epidural anaesthesia; the estimated RR was 0.17 (95% CI 0.02 to 1.44; 3 studies, 213 participants) in thoracic epidural anaesthesia.

The results of the planned sensitivity analyses showed minimal impact on the synthesised result. We performed only the following sensitivity analysis because we did not impute or transform the data for this outcome. See [Supplementary material 7](#).

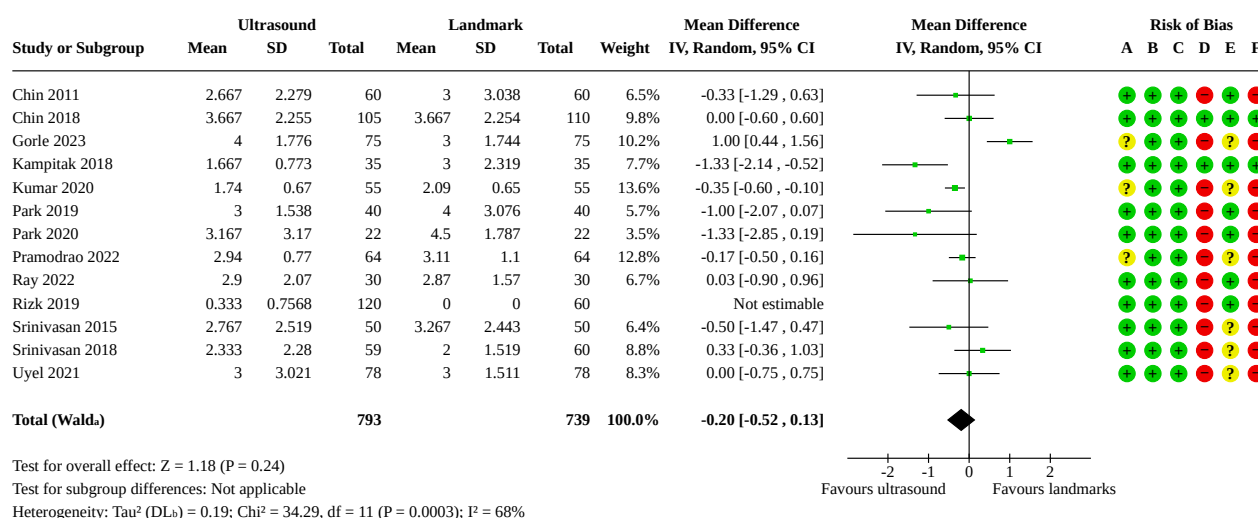
- Including only studies that were judged to be at overall low risk of bias (RR 1.05, 95% CI 0.58 to 1.92; $I^2 = 35\%$; 18 studies, 2216 participants).

Pain during procedure

Thirteen studies reported pain during the procedure. All studies used an 11-point numeric scale (0 means no pain) to measure pain.

Meta-analysis of the data from 1532 participants showed that ultrasound guidance compared with landmarks may result in little to no difference in pain during the procedure, but the evidence is very uncertain (MD -0.20 points, 95% CI -0.52 to 0.13; $I^2 = 68\%$; 13 studies, 1532 participants; very low-certainty evidence; Analysis 1.6, [Figure 8](#)). We downgraded the certainty of evidence for this outcome due to concerns about risk of bias, inconsistency, and imprecision.

Figure 8.



Footnotes

aCI calculated by Wald-type method.

bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

There were sufficient data to investigate three of the four expected effect modifiers with subgroup analyses. We could not examine the impact of types of ultrasound guidance, because all studies used pre-procedural ultrasound. The results did not suggest heterogeneity between the investigated subgroups.

- Difficult versus not difficult (test for subgroup differences: $\text{Chi}^2 = 1.50$, $\text{df} = 1$, $P = 0.22$, $I^2 = 33.4\%$)

- Experienced versus inexperienced (test for subgroup differences: $\text{Chi}^2 = 1.77$, $\text{df} = 1$, $P = 0.18$, $I^2 = 43.6\%$)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 0.39$, $\text{df} = 1$, $P = 0.53$, $I^2 = 0\%$)

The results of the planned sensitivity analyses showed minimal impact on the synthesised result. See [Supplementary material 7](#) for full analyses relating to subgroups and sensitivity analyses.

- Including only studies that were judged to be at overall low risk of bias (MD -0.64, 95% CI -1.94 to 0.67; $I^2 = 85\%$; 2 studies, 285 participants)
- Including only studies that reported both the mean and standard deviation (MD -0.27, 95% CI -0.46 to -0.08; $I^2 = 0\%$; 3 studies, 298 participants)

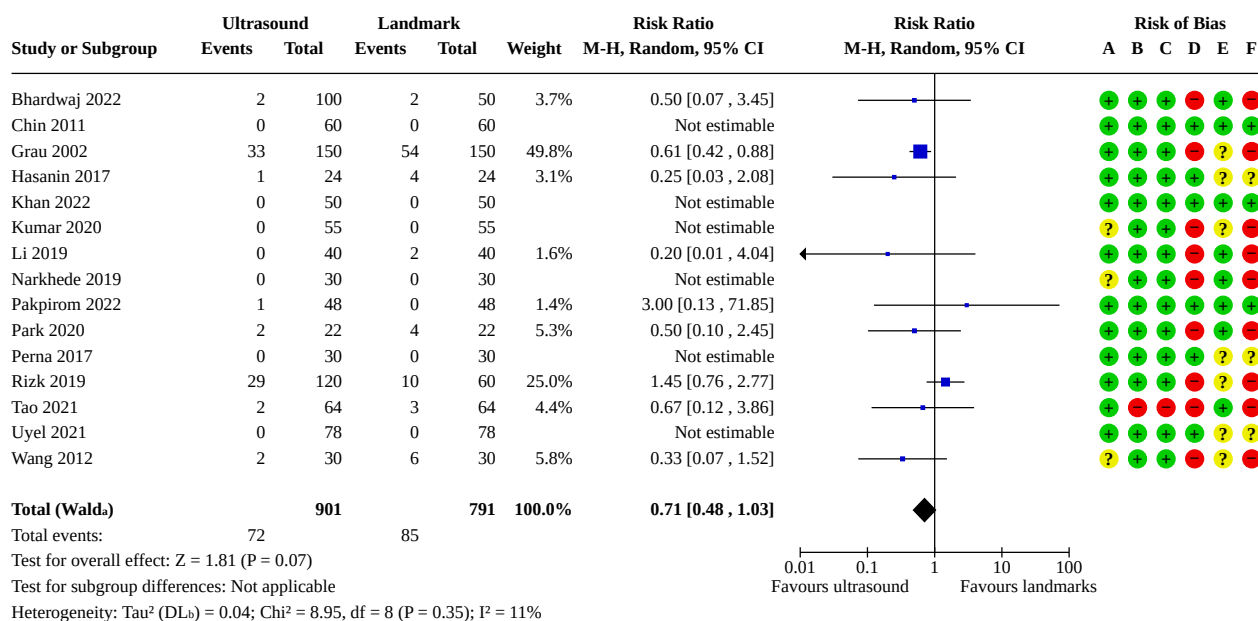
Any adverse events

Fifteen studies reported overall adverse event rates, i.e. the number of participants who experienced any complications. Thirty-five

studies reported only the incidence of each adverse event. The major frequent complication was bloody tap or paraesthesia. The details of those results are shown in [Table 2](#); [Table 3](#).

Meta-analysis of the data from 1692 participants showed that ultrasound guidance compared with palpation may result in little to no difference in risk of adverse events, but the evidence is very uncertain (RR 0.71, 95% CI 0.48 to 1.03; $I^2 = 11\%$; 15 studies, 1692 participants; very low-certainty evidence; Analysis 1.7, [Figure 9](#)). In six studies, the number of events was zero in both groups. We downgraded the certainty of evidence by one level for risk of bias, and two levels for imprecision.

Figure 9.



Footnotes

^aCI calculated by Wald-type method.

^bTau² calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

There were sufficient data to investigate the prespecified subgroup analyses.

- Difficult versus not difficult (test for subgroup differences: $\text{Chi}^2 = 0.08$, $df = 1$, $P = 0.78$, $I^2 = 0\%$)
- Experienced versus inexperienced (test for subgroup differences: $\text{Chi}^2 = 5.90$, $df = 1$, $P = 0.02$, $I^2 = 83.0\%$)
- Pre-procedural versus real-time techniques (test for subgroup differences: $\text{Chi}^2 = 0.10$, $df = 1$, $P = 0.75$, $I^2 = 0\%$)
- Spinal versus lumbar epidural versus thoracic epidural (test for subgroup differences: $\text{Chi}^2 = 1.19$, $df = 2$, $P = 0.55$, $I^2 = 0\%$)

The results suggest heterogeneity between the subgroups for the experience level of operators. For ultrasound versus landmarks, the estimated RR was 0.59 (95% CI 0.42 to 0.82; 9 studies, 1058 participants) for experienced operators; the estimated RR was 1.45 (95% CI 0.76 to 2.77; 2 studies, 290 participants) for inexperienced operators.

The planned sensitivity analysis (including only studies that were judged to be at overall low risk of bias) showed a very wide CI, because only one study was included in the analysis (RR 3.00, 95% CI 0.13 to 71.85; 1 study, 298 participants). See [Supplementary material 7](#).

Equity assessment

We did not examine any equity-related analysis in this review.

Reporting biases

We constructed a funnel plot to investigate potential publication bias. For the outcome, first-attempt success, the funnel plot was visually asymmetric (Figure 6). The Egger test also indicated the presence of funnel plot asymmetry ($P < 0.0001$). For other outcomes (number of attempts, procedure time, technical failure), the Egger test also indicated the presence of funnel plot asymmetry. However, we found reasonable visual symmetry in these funnel plots, so we did not downgrade the certainty of evidence for those outcomes due to publication bias.

DISCUSSION

Summary of main results

Cumulative data from 65 studies showed that, when compared to anatomical landmark palpation, ultrasound guidance for neuraxial anaesthesia reduces the number of attempts until success or procedure termination, and the procedure/needling time (high-certainty evidence). Ultrasound guidance likely increases the rate of first-attempt success (moderate-certainty evidence). Ultrasound guidance may result in little to no difference in participant satisfaction during the procedure, and technical failure of neuraxial anaesthesia (low-certainty evidence). Ultrasound guidance may result in little to no difference in pain during the procedure, and adverse events, but the evidence in both cases is very uncertain (very-low certainty evidence).

Limitations of the evidence included in the review

Overall, the included studies were of sufficient scope, and evaluated relevant outcome measures that addressed the objectives of our review.

About half of the included studies involved obstetric populations. They are more likely to have difficulty with neuraxial anaesthesia due to weight gain or oedema. For them, avoiding general anaesthesia is beneficial, and increasing the success rate of neuraxial anaesthesia is important. Most reasons for difficult neuraxial anaesthesia in the included studies were obesity and older age. The applicability of the results of our review may be limited for other populations for other reasons, such as previous spinal surgery. Our review included both pre-procedural and real-time ultrasound as interventions. In contrast, we excluded computer-aided, three-dimensional image neuraxial ultrasound, because this technique requires specialised ultrasound devices rather than conventional devices [104]. This newer guidance method should be evaluated in another review. Of the outcomes we prespecified, few studies reported on pain during the procedure. In addition, although many studies reported adverse events, few studies reported the number of participants who experienced any complications. For these outcomes, fewer studies were included in the meta-analysis, which may have reduced statistical power.

We downgraded the certainty of evidence for participant satisfaction, pain during the procedure, and adverse events, due to serious risk of bias. In those outcomes, the risk of bias mainly arose from the measurement of the outcome, since it was difficult to blind the outcome assessors to the intervention. Nearly half of the studies were judged to raise some concerns in the

domain, selection of the reported results, across all outcomes, because there were no publicly available trial registrations or study protocols. However, we did not downgrade the certainty of the evidence for this reason alone, as the absence of a publicly available protocol does not necessarily indicate selective reporting.

We carefully examined the studies for any inconsistencies between the methods and results sections, as well as for any evidence of planned but unreported outcomes. We downgraded the certainty of evidence for participant satisfaction and pain during the procedure due to inconsistency. In many outcomes, the I^2 and the χ^2 statistics indicated considerable heterogeneity, but it is worth noting that our review included many studies with small sample sizes, and thus the interpretation of heterogeneity should be taken with caution, as advised by the *Cochrane Handbook for Systematic Reviews of Interventions* [43]. We judged inconsistencies by both visual inspection of forest plots and statistical tests. We downgraded the certainty of evidence for pain during the procedure and adverse events due to imprecision. The wide range of confidence intervals and insufficient sample size contributed to this judgment. We downgraded the certainty of evidence for first-attempt success due to publication bias. Both the visual inspection of funnel plots and the Egger test indicated publication bias.

The studies included in this review may not fully reflect more recent, advanced devices, because some of the studies were published more than a decade ago. As ultrasound technology continues to evolve, with improvements in image resolution, software capabilities, availability of affordable devices, and device portability, the benefits of ultrasound guidance may become more pronounced and accessible in clinical practice.

In the context of applicability, ultrasound may be less likely to be implemented in low- and middle-income countries due to its cost. However, 39 of the included studies were conducted in middle-income countries, although none were conducted in low-income countries. Hence, the findings of this review may not be directly applicable to low-income settings. We did not perform additional subgroup analyses as we did not expect any differences in the effect of ultrasound between countries.

Limitations of the review processes

We made every effort to perform a highly sensitive search to identify potentially eligible studies through bibliographic databases and clinical trial databases. We encountered a limitation in the review process when attempting to collect additional information from study investigators. Despite our efforts to contact them for data clarification, we received few responses. Consequently, a relatively large number of studies were classified as 'awaiting classification'.

For continuous outcomes, many studies reported the results as median and interquartile range (IQR), or range. We converted those results to mean and standard deviation (SD), but we investigated the potential impact by sensitivity analyses. However, statistical heterogeneity may be present despite our pre-planned subgroup analyses. However, these subgroups were selected after careful consideration of clinically meaningful subpopulations, and we considered that the large number of studies included in our review made heterogeneity easier to detect.

We deviated from the study protocol to increase efficiency and facilitate interpretation of results. This included changing the

definition of the outcomes, number of attempts until success or procedure termination, and procedure time. For the former, it was mainly a name change from 'until pass success' to 'until success'. We refined the definition of 'procedure time' to 'needling time', which ultimately reduced the number of studies included in the data synthesis. See [Supplementary material 9](#) for further details of the differences between protocol and review.

None of the authors have any financial conflicts of interest related to this review topic, and none had any direct involvement in the conduct, analysis, or publication of the clinical trials related to this review.

Agreements and disagreements with other studies or reviews

Our results showed good agreement with similar previous reviews and meta-analyses regarding the number of attempts, first-attempt success, and technical failures [6, 17, 105], although no reviews addressed exactly the same scope as our question. In addition, reviews involving obstetric participants reported similar results to our review regarding first-attempt success and procedure time [19, 106]. The most recent reviews adopted network meta-analysis, which compares multiple interventions [26, 27, 28]. Two of the larger reviews are difficult to compare directly with our results because they included participants who underwent neuraxial puncture for other than anaesthetic purposes, and had different methodologies [26, 27]. However, it seemed that consistent results were obtained for the relevant outcomes. Compared to these existing reviews, we conducted a comprehensive search, and included grey literature, such as a Masters' thesis. Only a few previous reviews used participant-reported outcomes; our review included satisfaction, pain, and adverse events as outcomes.

AUTHORS' CONCLUSIONS

Implications for practice

We found that in adults undergoing neuraxial anaesthesia, ultrasound guidance reduces the number of attempts until success and also reduces procedure (needling) time when compared to anatomical landmarks; and it likely increases the rate of first-attempt success. Ultrasound guidance may result in little to no difference in participant satisfaction during procedure and technical failure of neuraxial anaesthesia. The evidence is very uncertain about the effects of ultrasound guidance on pain during procedure and adverse events.

Although ultrasound guidance appears to be generally beneficial for neuraxial anaesthesia, there are some uncertainties of evidence. The current evidence should be interpreted with caution when considering the use of ultrasound guidance for neuraxial anaesthesia in real-world practice.

Equity-related implications for practice

Discussed under [Limitations of the evidence included in the review](#).

Implications for research

Future studies could use a standardised definition of each outcome measure, especially measurement scales of satisfaction during the procedure. Participant-centred outcomes, such as satisfaction and pain, are important, and blinding of the participants, such as the use of sham ultrasound, is ideal for these outcome

measures. Although we understand the difficulties of blinding in this type of intervention, as shown in some studies included in our review, the implementation of sham ultrasound in participants in the control group was considered. We found that many studies were not registered in advance with a trial registry. Registration in a trial registry is mandatory prior to the start of a study. Furthermore, in our review, some characteristics, such as difficulty in palpating landmarks, were suggested as effect modifiers. Larger, well-designed randomised controlled trials, specific to these characteristics, may identify populations that may benefit from ultrasound guidance. Further research may be justified to investigate the differences between the effects of pre-procedural ultrasound and real-time ultrasound.

Equity-related implications for research

Discussed under [Limitations of the evidence included in the review](#).

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD014964](https://doi.org/10.1002/14651858.CD014964).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Difference between protocol and review

ADDITIONAL INFORMATION

Acknowledgements

Cochrane Anaesthesia supported the authors in the development of this review. The authors would like to thank Anne-Marie Klint Jørgensen (Information Specialist) and Naomi Dayan (Managing Editor) of Cochrane Anaesthesia for their helpful comments and guidance.

The following people conducted the editorial process for this review.

- Sign-off Editor (final editorial decision): Arash Afshari, Department of Pediatric and Obstetric Anesthesia, University of Copenhagen, Rigshospitalet, Denmark
- Managing Editors (selected peer reviewers, provided editorial guidance to authors, edited the article): Elizabeth Royle and Joey Kwong, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, supported editorial team): Lisa Wydrzynski, Cochrane Central Editorial Service

- Copy Editor (copy editing and production): Victoria Pennick, Cochrane Central Production Service
- Peer-reviewers (provided comments and recommended an editorial decision): Wuhua Ma, Department of Anesthesiology, The First Affiliated Hospital of Guangzhou University of Chinese Medicine (clinical/content review); José L. Arias-Buría, Department of Physical Therapy, Occupational Therapy, Rehabilitation and Physical Medicine, Universidad Rey Juan Carlos, Alcorcón, Spain (clinical/content review); Reem Fakak (consumer review); Clare Miles, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Cochrane Evidence Production and Methods Directorate (search review)

Contributions of authors

Screening search results: YM, KM, and DR

Full-text retrieval: YM, KM, SY, and DR

Searching reference lists: YM, KM, SY, and DR

Risk of bias assessment: YM, KM, SY, and DR

Data extraction: YM, KM, SY, and DR

Data entry and verification: YM, KM, SY, DR, and IN

Interpretation of data: YM, KM, SY, DR, IN, ES, and NW

Writing the review: YM, KM, SY, DR, IN, ES, and NW

Final approval of the document to be published: YM, KM, SY, DR, IN, ES, and NW

Declarations of interest

Yuto Makino: none known

Kentaro Miyake: none known

David Roche: no relevant interests; Roche D, Iohom G. Point-of-care Ultrasound in Anaesthesia and Intensive Care Medicine. *Romanian Journal of Anaesthesia and Intensive Care*. 2018;25(2):95-96. doi: 10.21454/rjaic.7518.252.ioh; works as a specialist anaesthetist

Satoshi Yoshimura: no relevant interests; works as a physician

History

Protocol first published: Issue 8, 2022

Isao Nahara: none known

Ethan Sahker: Japanese Society for the Promotion of Science (Grant/Contract)

Norio Watanabe: no relevant interests; Senior Editor, Cochrane Database of Systematic Reviews, and was not involved in the decision-making or editorial processing of this review.

Sources of support

Internal sources

- none, Other

no internal sources of support received

External sources

- none, Other

no external sources of support received

Registration and protocol

Protocol (2023) DOI: 10.1002/14651858.CD014964

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library ([Supplementary material 8](#)): full search strategies for each database ([Supplementary material 1](#)); full citations of each unique report for all studies included ([Supplementary material 2](#)), ongoing ([Supplementary material 5](#)) or awaiting classification ([Supplementary material 4](#)), or excluded at the full text screen ([Supplementary material 3](#)) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments ([Supplementary material 6](#)); and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows ([Supplementary material 7](#)). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods. Template data extraction forms from Covidence are available from the authors on reasonable request.

REFERENCES

1. Li J, Krishna R, Zhang Y, Lam D, Vadivelu N. Ultrasound-guided neuraxial anesthesia. *Current Pain and Headache Reports* 2020;**24**(10):59. [PMID: 32812182]
2. De Filho GR, Gomes HP, Da Fonseca MH, Hoffman JC, Pederneiras SG, Garcia JH. Predictors of successful neuraxial block: a prospective study. *European Journal of Anaesthesiology* 2002;**19**(6):447-51. [PMID: 12094920]
3. Parra MC, Washburn K, Brown JR, Beach ML, Yeager MP, Barr P, et al. Fluoroscopic guidance increases the incidence of thoracic epidural catheter placement within the epidural space: a randomized trial. *Regional Anesthesia and Pain Medicine* 2017;**42**(1):17-24. [PMID: 27922948]
4. Broadbent CR, Maxwell WB, Ferrie R, Wilson DJ, Gawne-Cain M, Russell R. Ability of anaesthetists to identify a marked lumbar interspace. *Anaesthesia* 2000;**55**(11):1122-6. [PMID: 11069342]
5. Furness G, Reilly MP, Kuchi S. An evaluation of ultrasound imaging for identification of lumbar intervertebral level. *Anaesthesia* 2002;**57**(3):277-80. [PMID: 11892638]
6. Perlas A, Chaparro LE, Chin KJ. Lumbar neuraxial ultrasound for spinal and epidural anesthesia: a systematic review and meta-analysis. *Regional Anesthesia and Pain Medicine* 2016;**41**(2):251-60. [PMID: 25493689]
7. Bos EM, Haumann J, de Quelerij M, Vandertop WP, Kalkman CJ, Hollmann MW, et al. Haematoma and abscess after neuraxial anaesthesia: a review of 647 cases. *British Journal of Anaesthesia* 2018;**120**(4):693-704. [PMID: 29576110]
8. Bogin IN, Stulin ID. Application of the method of 2-dimensional echospondylography for determining landmarks in lumbar punctures. *Zhurnal Nevropatologii i Psikiatrii Imeni S.S. Korsakova* 1971;**71**(12):1810-1. [PMID: 4335474]
9. Cork RC, Kryc JJ, Vaughan RW. Ultrasonic localization of the lumbar epidural space. *Anesthesiology* 1980;**52**(6):513-6. [PMID: 7377598]
10. Grau T, Leipold RW, Horter J, Conradi R, Martin E, Motsch J. The lumbar epidural space in pregnancy: visualization by ultrasonography. *British Journal of Anaesthesia* 2001;**86**(6):798-804. [PMID: 11573586]
11. Grau T, Leipold RW, Conradi R, Martin E, Motsch J. The visualisation of dura perforation and blood patches with ultrasound. *Anesthesiologie, Intensivmedizin, Notfallmedizin, Schmerztherapie* 2002;**37**(3):149-53. [PMID: 11889616]
12. Karmakar MK, Li X, Ho AM, Kwok WH, Chui PT. Real-time ultrasound-guided paramedian epidural access: evaluation of a novel in-plane technique. *British Journal of Anaesthesia* 2009;**102**(6):845-54. [PMID: 19398454]
13. Chin KJ. Recent developments in ultrasound imaging for neuraxial blockade. *Current Opinion in Anaesthesiology* 2018;**31**(5):608-13. [PMID: 29994941]
14. Guay J, Suresh S, Kopp S. The use of ultrasound guidance for perioperative neuraxial and peripheral nerve blocks in children. *Cochrane Database of Systematic Reviews* 2019, Issue 2. Art. No: CD011436. [DOI: [10.1002/14651858.CD011436.pub3](https://doi.org/10.1002/14651858.CD011436.pub3)]
15. Kim WJ, Kim TH, Shin HY, Kang H, Baek CW, Jung YH, et al. Fluoroscope guided epidural needle insertion in midthoracic region: clinical evaluation of Nagaro's method. *Korean Journal of Anesthesiology* 2012;**62**(5):441-7. [PMID: 22679541]
16. Lewis SR, Price A, Walker KJ, McGrattan K, Smith AF. Ultrasound guidance for upper and lower limb blocks. *Cochrane Database of Systematic Reviews* 2015, Issue 9. Art. No: CD006459. [DOI: [10.1002/14651858.CD006459.pub3](https://doi.org/10.1002/14651858.CD006459.pub3)]
17. Shaikh F, Brzezinski J, Alexander S, Arzola C, Carvalho JC, Beyene J, et al. Ultrasound imaging for lumbar punctures and epidural catheterisations: systematic review and meta-analysis. *BMJ* 2013;**346**:f1720. [PMID: 23532866]
18. Sidiropoulou T, Christodoulaki K, Siristatidis C. Pre-procedural lumbar neuraxial ultrasound – a systematic review of randomized controlled trials and meta-analysis. *Healthcare (Basel, Switzerland)* 2021;**9**(4):479. [PMID: 33920621]
19. Jiang L, Zhang F, Wei N, Lv J, Chen W, Dai Z. Could preprocedural ultrasound increase the first-pass success rate of neuraxial anesthesia in obstetrics? A systematic review and meta-analysis of randomized controlled trials. *Journal of Anesthesia* 2020;**34**(3):434-44. [PMID: 32133540]
20. Li M, Ni X, Xu Z, Shen F, Song Y, Li Q, et al. Ultrasound-assisted technology versus the conventional landmark location method in spinal anesthesia for cesarean delivery in obese parturients: a randomized controlled trial. *Anesthesia and Analgesia* 2019;**129**(1):155-61. [DOI: [10.1213/ANE.0000000000003795](https://doi.org/10.1213/ANE.0000000000003795)]
21. Park SK, Yoo S, Kim WH, Lim YJ, Bahk JH, Kim JT. Ultrasound-assisted vs. landmark-guided paramedian spinal anaesthesia in the elderly: a randomised controlled trial. *European Journal of Anaesthesiology* 2019;**36**(10):763-71. [DOI: [10.1097/EJA.0000000000001029](https://doi.org/10.1097/EJA.0000000000001029)]
22. Arzola C, Mikhael R, Margarido C, Carvalho JC. Spinal ultrasound versus palpation for epidural catheter insertion in labour. *European Journal of Anaesthesiology* 2015;**32**(7):499-505. [DOI: [10.1097/EJA.0000000000000119](https://doi.org/10.1097/EJA.0000000000000119)]
23. Turkstra TP, Marmai KL, Armstrong KP, Kumar K, Singh SI. Preprocedural ultrasound assessment does not improve trainee performance of spinal anesthesia for obstetrical patients: a randomized controlled trial. *Journal of Clinical Anesthesia* 2017;**37**:21-4. [DOI: [10.1016/j.jclinane.2016.10.034](https://doi.org/10.1016/j.jclinane.2016.10.034)]
24. Chong SE, Mohd Nikman A, Saedah A, Wan Mohd Nazaruddin WH, Kueh YC, Lim JA, et al. Real-time ultrasound-guided paramedian spinal anaesthesia: evaluation of the efficacy and the success rate of single needle pass. *British Journal of Anaesthesia* 2017;**118**(5):799-801. [PMID: 28510752]

25. Elsharkawy H, Maheshwari A, Babazade R, Perlas A, Zaky S, Mounir-Soliman L. Real-time ultrasound-guided spinal anesthesia in patients with predicted difficult anatomy. *Minerva Anestesiologica* 2017;**83**(5):465-73. [DOI: [10.23736/S0375-9393.16.11610-4](https://doi.org/10.23736/S0375-9393.16.11610-4)]
26. De Carvalho CC, Genuino WP, Morais MC, de Paiva Oliveira H, Rodrigues AI, El-Boghdady K. Efficacy and safety of ultrasound-guided versus landmark-guided neuraxial puncture: a systematic review, network meta-analysis and trial sequential analysis of randomized clinical trials. *Regional Anesthesia and Pain Medicine* 2024 Jun 14 [Epub ahead of print]. [DOI: [10.1136/rapm-2024-105547](https://doi.org/10.1136/rapm-2024-105547)]
27. Kamimura Y, Yamamoto N, Shiroshita A, Miura T, Tsuji T, Someko H, et al. Comparative efficacy of ultrasound guidance or conventional anatomical landmarks for neuraxial puncture in adult patients: a systematic review and network meta-analysis. *British Journal of Anaesthesia* 2024;**132**(5):1097-111.
28. Zhang Y, Peng M, Wei J, Huang J, Ma W, Li Y. Comparison of ultrasound-guided and traditional localisation in intraspinal anesthesia: a systematic review and network meta-analysis. *BMJ Open* 2023;**13**(11):e071253.
29. Wynd KP, Smith HM, Jacob AK, Torsher LC, Kopp SL, Hebl JR. Ultrasound machine comparison: an evaluation of ergonomic design, data management, ease of use, and image quality. *Regional Anesthesia and Pain Medicine* 2009;**34**(4):349-56. [PMID: 19574868]
30. Higgins JPT, Lasserson T, Thomas J, Flemyng E, Churchill R. Standards for the conduct of new Cochrane intervention reviews. In: Higgins JPT, Lasserson T, Thomas J, Flemyng E, Churchill R. *Methodological Expectations of Cochrane Intervention Reviews*. Cochrane, London, August 2023. Available from <https://community.cochrane.org/mecir-manual>.
31. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]
32. Sahin T, Balaban O, Sahin L, Solak M, Toker K. A randomized controlled trial of preinsertion ultrasound guidance for spinal anaesthesia in pregnancy: outcomes among obese and lean parturients - ultrasound for spinal anesthesia in pregnancy. *Journal of Anesthesia* 2014;**28**(3):413-9. [DOI: [10.1007/s00540-013-1726-1](https://doi.org/10.1007/s00540-013-1726-1)]
33. Ansari T, Yousef A, El Gamassy A, Fayed M. Ultrasound-guided spinal anaesthesia in obstetrics: is there an advantage over the landmark technique in patients with easily palpable spines? *International Journal of Obstetric Anesthesia* 2014;**23**(3):213-6. [DOI: [10.1016/j.ijoa.2014.03.001](https://doi.org/10.1016/j.ijoa.2014.03.001)]
34. Glanville J, Briscoe S, Featherstone R, Littlewood A, Metzendorf M-I, Noel-Storr A, et al. Chapter 4: Searching for and selecting studies. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated October 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
35. Covidence. Melbourne, Australia: Veritas Health Innovation, accessed 20 June 2024. Available at <https://www.covidence.org>.
36. Review Manager (RevMan). Version 8.1.1. The Cochrane Collaboration, 2024. Available at <https://revman.cochrane.org>.
37. Higgins JPT, Savović J, Page MJ, Elbers RG, Sterne JAC. Chapter 8: Assessing risk of bias in a randomized trial. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
38. Higgins JPT, Eldridge S, Li T. Chapter 23: Including variants on randomized trials (last updated October 2019). In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
39. Cohen J. *Statistical Power Analysis in the Behavioral Sciences*. 2nd edition. Hillsdale (NJ): Lawrence Erlbaum Associates, Inc, 1988.
40. Higgins JPT, Li T, Deeks JJ, editor(s). Chapter 6: Choosing effect measures and computing estimates of effect. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
41. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Medical Research Methodology* 2014;**14**:135.
42. Egger M, Davey Smith G, Schneider M, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;**315**:629-34.
43. Deeks JJ, Higgins JP, Altman DG, editor(s). Chapter 10: Analysing data and undertaking meta-analyses. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
44. Schünemann HJ, Higgins JPT, Vist GE, Glasziou P, Akl EA, Skoetz N, et al. Chapter 14: Completing 'Summary of findings' tables and grading the certainty of the evidence. In: Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al, editor(s). *Cochrane Handbook for Systematic Reviews of Interventions* Version 6.4 (updated August 2023). Cochrane, 2023. Available from www.training.cochrane.org/handbook.
45. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *Journal of Clinical Epidemiology* 2011;**64**(4):383-94. [DOI: [10.1016/j.jclinepi.2010.04.026](https://doi.org/10.1016/j.jclinepi.2010.04.026)] [PMID: 21195583]
46. Bal AC, Singh S, Eapen S, Gupta A, Taank P. To study efficacy of ultrasound guided paramedian technique in administering

spinal anesthesia. *International Journal of Medical and Health Research* 2019;**5**(8):128-30.

47. Balaban O, Sahin T, Sahin L, Solak M, Toker K. Does lumbarultrasonography improve epidural catheterization for laboranalgesia? A randomized controlled study. *Middle East Journalof Anesthesiology* 2017;**24**(2):89-96.

48. Vernon TJ, Vogel TM, Dalby PL, Mandell G, Lim G. Ultrasound-assisted epidural labor analgesia for landmark identification in morbidly obese pregnant women: A preliminary investigation. *Journal of Clinical Anesthesia* 2020;**59**:53-4. [DOI: [10.1016/j.jclinane.2019.05.023](https://doi.org/10.1016/j.jclinane.2019.05.023)]

49. Wilkes D, Martinello C, Medeiros FA, Babazade R, Hurwitz E, Khanjee N, et al. Ultrasound-determined landmarks decrease pressure pain at epidural insertion site in immediate post-partum period. *Minerva Anestesiologica* 2017;**83**(10):1034-41. [DOI: [10.23736/S0375-9393.17.11782-7](https://doi.org/10.23736/S0375-9393.17.11782-7)]

50. Abdelhamid SA, Mansour MA. Ultrasound-guided intrathecal anesthesia: does scanning help? *Egyptian Journal of Anaesthesia* 2013;**29**(4):389-94. [DOI: [10.1016/j.egja.2013.06.003](https://doi.org/10.1016/j.egja.2013.06.003)]

51. Arzola C, Balki M, Gleicher Y, Malavade A, Friedman Z. Comparison of ultrasound-assistance versus traditional palpation method for placement of thoracic epidural catheters: a randomized controlled trial. *Regional Anesthesia and Pain Medicine* 2022;**47**(9):571-2. [DOI: [10.1136/rapm-2021-103296](https://doi.org/10.1136/rapm-2021-103296)]

52. Auyong DB, Hostetter L, Yuan SC, Slee AE, Hanson NA. Evaluation of ultrasound-assisted thoracic epidural placement in patients undergoing upper abdominal and thoracic surgery: a randomized, double-blind study. *Regional Anesthesia and Pain Medicine* 2017;**42**(2):204-9. [DOI: [10.1097/AAP.0000000000000540](https://doi.org/10.1097/AAP.0000000000000540)]

53. Bhardwaj D, Thakur L, Sharma S, Rana S, Gupta B, Sharma C. Comparative evaluation of three techniques for paramedian subarachnoid block: point-of-care preprocedural ultrasound assisted, real-time ultrasound guided and landmark based. *Indian Journal of Anaesthesia* 2022;**66**(Suppl 2):S102-7. [DOI: [10.4103/ija.ija_373_21](https://doi.org/10.4103/ija.ija_373_21)]

54. Chin KJ, Perlas A, Chan V, Brown-Shreves D, Koshkin A, Vaishnav V. Ultrasound imaging facilitates spinal anesthesia in adults with difficult surface anatomic landmarks. *Anesthesiology* 2011;**115**(1):94-101. [DOI: [10.1097/ALN.0b013e31821a8ad4](https://doi.org/10.1097/ALN.0b013e31821a8ad4)]

55. Chin A, Crooke B, Heywood L, Brijball R, Pelecanos AM, Abeyapala W. A randomised controlled trial comparing needle movements during combined spinal-epidural anaesthesia with and without ultrasound assistance. *Anaesthesia* 2018;**73**(4):466-73. [DOI: [10.1111/anae.14206](https://doi.org/10.1111/anae.14206)]

56. Creaney M, Mullane D, Casby C, Tan T. Ultrasound to identify the lumbar space in women with impalpable bony landmarks presenting for elective caesarean delivery under spinal anaesthesia: a randomised trial. *International Journal of Obstetric Anaesthesia* 2016;**28**:12-6. [DOI: [10.1016/j.ijoa.2016.07.007](https://doi.org/10.1016/j.ijoa.2016.07.007)]

57. Dhanger S, Vinayagam S, Vaidhyanathan B, Rajesh IJ, Tripathy DK. Comparison of landmark versus pre-procedural ultrasonography-assisted midline approach for identification of subarachnoid space in elective caesarean section: a randomised controlled trial. *Indian Journal of Anaesthesia* 2018;**62**(4):280-4. [DOI: [10.4103/ija.IJA_488_17](https://doi.org/10.4103/ija.IJA_488_17)]

58. Ekinci M, Alici HA, Ahiskalioglu A, Ince I, Aksoy M, Celik EC, et al. The use of ultrasound in planned cesarean delivery under spinal anesthesia for patients having nonprominent anatomic landmarks. *Journal of Clinical Anesthesia* 2017;**37**:82-5. [DOI: [10.1016/j.jclinane.2016.10.014](https://doi.org/10.1016/j.jclinane.2016.10.014)]

59. Galal SM, Mohamed DZ, Fouad AZ. Ultrasound guidance for introducing spinal anesthesia through paramedian approach versus landmark guided technique during lower limb surgeries, randomized controlled trial. *Indian Journal of Public Health Research and Development* 2019;**10**(12):1289-92. [DOI: [10.37506/v10/i12/2019/ijphrd/192225](https://doi.org/10.37506/v10/i12/2019/ijphrd/192225)]

60. Gayathri B, Swetha Ramani CK, Urkavalan K, Pushparani A, Rajendran A. Comparison of the time taken for subarachnoid block using ultrasound-guided method versus landmark technique for cesarean section - a randomized controlled study. *Journal of Anaesthesiology, Clinical Pharmacology* 2021;**37**(2):205-9. [DOI: [10.4103/joacp.JOACP_35_20](https://doi.org/10.4103/joacp.JOACP_35_20)]

61. Geng J, Chen X, Wang X, Guo X, Li M. Ultrasound imaging increases first-attempt success rate of neuraxial block in elderly patients. *National Medical Journal of China* 2016;**96**(43):3459-63. [DOI: [10.3760/cma.j.issn.0376-2491.2016.43.004](https://doi.org/10.3760/cma.j.issn.0376-2491.2016.43.004)]

62. Gorle Y, Papireddy SM, Tarigonda S. Evaluation of lumbar sonography as a learning aid for performing subarachnoid block using the paramedian approach by medical junior resident anaesthesiologists: a randomized controlled trial. *Cureus* 2023;**15**(5):e38871. [DOI: [10.7759/cureus.38871](https://doi.org/10.7759/cureus.38871)]

63. Grau T, Leipold RW, Conradi R, Martin E. Ultrasound control for presumed difficult epidural puncture. *Acta Anaesthesiologica Scandinavica* 2001;**45**(6):766-71. [DOI: [10.1034/j.1399-6576.2001.045006766.x](https://doi.org/10.1034/j.1399-6576.2001.045006766.x)]

64. Grau T, Leipold RW, Conradi R, Martin E, Motsch J. Ultrasound imaging facilitates localization of the epidural space during combined spinal and epidural anesthesia. *Regional Anesthesia and Pain Medicine* 2001;**26**(1):64-7. [DOI: [10.1053/rapm.2001.19633](https://doi.org/10.1053/rapm.2001.19633)]

65. Grau T, Leipold RW, Conradi R, Martin E, Motsch J. Efficacy of ultrasound imaging in obstetric epidural anesthesia. *Journal of Clinical Anesthesia* 2002;**14**(3):169-75. [DOI: [10.1016/S0952-8180%2801%2900378-6](https://doi.org/10.1016/S0952-8180%2801%2900378-6)]

66. Grau T, Leipold RW, Fatehi S, Martin E, Motsch J. Real-time ultrasonic observation of combined spinal-epidural anaesthesia. *European Journal of Anaesthesiology* 2004;**21**(1):25-31. [DOI: [10.1017/S026502150400105X](https://doi.org/10.1017/S026502150400105X)]

67. Hasanin AM, Mokhtar AM, Amin SM, Sayed AA. Preprocedural ultrasound examination versus manual palpation for thoracic epidural catheter insertion. *Saudi Journal of Anaesthesia* 2017;**11**(1):62-6. [DOI: [10.4103/1658-354X.197345](https://doi.org/10.4103/1658-354X.197345)]

68. Isiker A, Onay M, Akcan A, Akkemik U, Gulec MS. Evaluating the role of vertebral anatomy examination by ultrasonography before administering spinal anesthesia in geriatric patients: a prospective randomized trial. *Turkish Journal of Geriatrics* 2023;**26**(1):79-90. [DOI: [10.29400/tjgeri.2023.333](https://doi.org/10.29400/tjgeri.2023.333)]
69. Jagadish A, Swaminathan S, Bidkar PU, Gupta SL, Adinarayanan S. Ease of lumbar epidural catheter insertion with prepuncture ultrasound as guidance compared with conventional palpatory technique when performed by anesthesiology residents: a randomized controlled trial. *Journal of Anaesthesiology, Clinical Pharmacology* 2021;**37**(2):216-20. [DOI: [10.4103/joacp.JOACP_394_19](https://doi.org/10.4103/joacp.JOACP_394_19)]
70. Jain K, Puri A, Taneja R, Jaiswal V, Jain A. Preprocedural ultrasound as an adjunct to blind conventional technique for epidural neuraxial blockade in patients undergoing hip or knee joint replacement surgery: a randomised controlled trial. *Indian Journal of Anaesthesia* 2019;**63**(11):924-31. [DOI: [10.4103/ija.IJA_327_19](https://doi.org/10.4103/ija.IJA_327_19)]
71. Jayanth MN, Arumulla SP, Kesana P, Kandukuru KC, Basireddy HR, Peddi S. Preprocedural ultrasonography as an adjunct to landmark technique for identification of epidural space in parturients for labor analgesia. *Saudi Journal of Anaesthesia* 2023;**17**(1):18-22. [DOI: [10.4103/sja.sja_141_22](https://doi.org/10.4103/sja.sja_141_22)]
72. Jiang J, Xu Y, He M. Real-time ultrasound-guided vs. anatomical landmark-guided paramedian epidural anesthesia in overweight parturients undergoing analgesic labor: a randomized controlled trial. *Signa Vitae* 2021;**17**(6):66-73. [DOI: [10.22514/sv.2021.218](https://doi.org/10.22514/sv.2021.218)]
73. Kampitak W, Werawatganon T, Uerpairojkit K, Songthamwat B. Paramedian spinal anesthesia: landmark vs. ultrasound-guided approaches. *Journal of Anesthesia and Clinical Research* 2018;**9**(6):100837. [DOI: [10.4172/2155-6148.1000837](https://doi.org/10.4172/2155-6148.1000837)]
74. Kawaguchi RI, Yamauchi M, Sugino S, Yamakage M. Ultrasound-aided ipsilateral-dominant epidural block for total hip arthroplasty: a randomised controlled single-blind study. *European Journal of Anaesthesiology* 2011;**28**(2):137-40. [DOI: [10.1097/EJA.0b013e3283423457](https://doi.org/10.1097/EJA.0b013e3283423457)]
75. Khan MA, Gupta M, Sharma S, Kasaudhan S. A comparative study of ultrasound assisted versus landmark technique for combined spinal-epidural anaesthesia in patients undergoing lower limb orthopaedic surgery. *Indian Journal of Anaesthesia* 2022;**66**(4):272-7. [DOI: [10.4103/ija.ija_775_21](https://doi.org/10.4103/ija.ija_775_21)]
76. Korda B, Scribante J, Perrie H, Nel D. Ultrasound-assisted versus landmark-based spinal block performance in emergency caesarean delivery in obese patients at a central hospital [Masters thesis]. Faculty of Health Sciences, University of the Witwatersrand, Johannesburg. 2020. Available at <https://wiredspace.wits.ac.za/server/api/core/bitstreams/7852a54e-d0f6-451a-8d30-e6e576626205/content>.
77. Kumar A, Thakur P, Wason R. Ultrasound guided paramedian versus landmark guided midline technique for spinal anaesthesia-a randomised clinical trial. *Journal of Clinical and Diagnostic Research* 2020;**14**(12):5-9. [DOI: [10.7860/JCDR/2020/45793.14367](https://doi.org/10.7860/JCDR/2020/45793.14367)]
78. Lim YC, Choo CY, Tan KT. A randomised control trial on ultrasound-assisted spinal anaesthesia. *Regional Anesthesia and Pain Medicine* 2012;**37**(5):E258. [DOI: [10.1097/AAP.0b013e31826a8366](https://doi.org/10.1097/AAP.0b013e31826a8366)]
79. Narkhede H, Kane D, Parekh V, Hemantkumar I. A cohort study of anatomical landmark-guided midline versus pre-procedure ultrasound-guided midline technique of spinal anesthesia in elderly patients undergoing orthopedic surgery. *Journal of Anaesthesiology, Clinical Pharmacology* 2019;**35**(4):522-7. [DOI: [10.4103/joacp.JOACP_396_17](https://doi.org/10.4103/joacp.JOACP_396_17)]
80. Nassar M, Abdelazim IA. Pre-puncture ultrasound guided epidural insertion before vaginal delivery. *Journal of Clinical Monitoring and Computing* 2015;**29**(5):573-7. [DOI: [10.1007/s10877-014-9634-y](https://doi.org/10.1007/s10877-014-9634-y)]
81. NCT02207972. Does ultrasound help with placement of labor analgesia in pregnant patients? <https://clinicaltrials.gov/study/NCT02207972?id=NCT02207972&rank=1> (first posted 01 August 2014).
82. NCT02553746. Ultrasound for neuraxial anesthesia. <https://clinicaltrials.gov/study/NCT02553746?id=NCT02553746&rank=1> (first posted 16 September 2015).
83. Pakpirom J, Thatsanapornsathit K, Kovitwanawong N, Petsakul S, Benjhawaleemas P, Narunart K, et al. Real-time ultrasound-guided versus anatomic landmark-based thoracic epidural placement: a prospective, randomized, superiority trial. *BMC Anesthesiology* 2022;**22**(1):198. [DOI: [10.1186/s12871-022-01730-5](https://doi.org/10.1186/s12871-022-01730-5)]
84. Park SK, Bae J, Yoo S, Kim WH, Lim YJ, Bahk JH, et al. Ultrasound-assisted versus landmark-guided spinal anesthesia in patients with abnormal spinal anatomy: a randomized controlled trial. *Anesthesia and Analgesia* 2020;**130**(3):787-95. [DOI: [10.1213/ANE.0000000000004600](https://doi.org/10.1213/ANE.0000000000004600)]
85. Perna P, Gioia A, Ragazzi R, Volta CA, Innamorato M. Can pre-procedure neuroaxial ultrasound improve the identification of the potential epidural space when compared with anatomical landmarks? A prospective randomized study. *Minerva Anestesiologica* 2017;**83**(1):41-9. [DOI: [10.23736/S0375-9393.16.11399-9](https://doi.org/10.23736/S0375-9393.16.11399-9)]
86. Pramodrao BN, Sharma VK, Mishra S, Nimje SV. To find out the difference in the number of passes between landmark-guided midline and preprocedure ultrasound-guided paramedian techniques in spinal anesthesia. *Journal of Cardiovascular Disease Research* 2022;**13**(8):3449-55. [DOI: [10.31838/jcdr.2022.13.08.453](https://doi.org/10.31838/jcdr.2022.13.08.453)]
87. Qu B, Chen L, Zhang Y, Jiang M, Wu C, Ma W, et al. Landmark-guided versus modified ultrasound-assisted paramedian techniques in combined spinal-epidural anesthesia for elderly patients with hip fractures: a randomized controlled trial. *BMC Anesthesiology* 2020;**20**(1):248. [DOI: [10.1186/s12871-020-01172-x](https://doi.org/10.1186/s12871-020-01172-x)]
88. Rahimzadeh P, Ghodraty MR, Faiz SH, Pardis E, Yavari T, Pournajafian A. Success rate of spinal anesthesia by anesthesia resident: comparing ultrasonography versus traditional method, a randomized clinical trial. *Journal of Kerman*

University of Medical Sciences 2022;**29**(3):218-25. [DOI: [10.22062/jkmu.2022.91945](https://doi.org/10.22062/jkmu.2022.91945)]

89. Ray T, Saxena S, Panda A. Preprocedural ultrasound assessment of landmarks of paramedian approach for subarachnoid block in geriatric population. *Indian Anaesthetists Forum* 2022;**23**(1):38-43. [DOI: [10.4103/TheIAForum.TheIAForum_97_21](https://doi.org/10.4103/TheIAForum.TheIAForum_97_21)]

90. Rizk MS, Zeeni CA, Bouez JN, Bteich NJ, Sayyid SK, Alfahel WS, et al. Preprocedural ultrasound versus landmark techniques for spinal anesthesia performed by novice residents in elderly: a randomized controlled trial. *BMC Anesthesiology* 2019;**19**(1):208. [DOI: [10.1186/s12871-019-0882-8](https://doi.org/10.1186/s12871-019-0882-8)]

91. Singh S, Ravi PR, Mishra A. Comparison between the efficacy of preprocedural ultrasound-guided paramedian approach & conventional landmark guided paramedian approach in an obese patient. *International Journal of Pharmaceutical and Clinical Research* 2023;**15**(9):141-51.

92. Srinivasan KK, Iohom G, Loughnane F, Lee PJ. Conventional landmark-guided midline versus preprocedure ultrasound-guided paramedian techniques in spinal anesthesia. *Anesthesia and Analgesia* 2015;**121**(4):1089-96. [DOI: [10.1213/ANE.0000000000000911](https://doi.org/10.1213/ANE.0000000000000911)]

93. Srinivasan KK, Leo AM, Iohom G, Loughnane F, Lee PJ. Pre-procedure ultrasound-guided paramedian spinal anaesthesia at L5-S1: is this better than landmark-guided midline approach? A randomised controlled trial. *Indian Journal of Anaesthesia* 2018;**62**(1):53-60. [DOI: [10.4103/ija.IJA_448_17](https://doi.org/10.4103/ija.IJA_448_17)]

94. Tao B, Liu K, Ding M, Xue H, Li X, Zhao P. Ultrasound increases the success rate of spinal needle placement through the epidural needle during combined spinal-epidural anaesthesia: a randomised controlled study. *European Journal of Anaesthesiology* 2021;**38**(3):251-8. [DOI: [10.1097/EJA.0000000000001380](https://doi.org/10.1097/EJA.0000000000001380)]

95. Tawfik MM, Atallah MM, Elkharrboutly WS, Allakkany NS, Abdelkhalek M. Does preprocedural ultrasound increase the first-pass success rate of epidural catheterization before cesarean delivery? A randomized controlled trial. *Anesthesia and Analgesia* 2017;**124**(3):851-6. [DOI: [10.1213/ANE.0000000000001325](https://doi.org/10.1213/ANE.0000000000001325)]

96. Tawfik MM, Tolba MA, Ismail OM, Messeha MM. Ultrasonography versus palpation for spinal anesthesia in obese parturients undergoing cesarean delivery: a randomized controlled trial. *Regional Anesthesia and Pain Medicine* 2024;**49**(1):41-8. [DOI: [10.1136/rapm-2022-104272](https://doi.org/10.1136/rapm-2022-104272)]

97. Tubinis MD, Lester SA, Schlitz CN, Morgan CJ, Sakawi Y, Powell MF. Utility of ultrasonography in identification of

midline and epidural placement in severely obese parturients. *Minerva Anestesiologica* 2019;**85**(10):1089-96. [DOI: [10.23736/S0375-9393.19.13617-6](https://doi.org/10.23736/S0375-9393.19.13617-6)]

98. Urfalioglu A, Bilal B, Oksuz G, Bakacak M, Boran OF, Oksuz H. Comparison of the landmark and ultrasound methods in cesarean sections performed under spinal anesthesia on obese pregnant. *Journal of Maternal-Fetal & Neonatal Medicine* 2017;**30**(9):1051-6. [DOI: [10.1080/14767058.2016.1199677](https://doi.org/10.1080/14767058.2016.1199677)]

99. Uyel Y, Kilicaslan A. Preprocedural ultrasonography versus landmark-guided spinal anesthesia in geriatric patients with difficult anatomy: a prospective randomized trial. *Eurasian Journal of Medicine* 2021;**53**(1):9-14. [DOI: [10.5152/eurasianjmed.2020.20215](https://doi.org/10.5152/eurasianjmed.2020.20215)]

100. Vallejo MC, Phelps AL, Singh S, Orebaugh SL, Sah N. Ultrasound decreases the failed labor epidural rate in resident trainees. *International Journal of Obstetric Anesthesia* 2010;**19**(4):373-8. [DOI: [10.1016/j.ijoa.2010.04.002](https://doi.org/10.1016/j.ijoa.2010.04.002)]

101. Wang Q, Yin C, Wang TL. Ultrasound facilitates identification of combined spinal-epidural puncture in obese parturients. *Chinese Medical Journal* 2012;**125**(21):3840-3. [DOI: [10.3760/cma.j.issn.0366-6999.2012.21.018](https://doi.org/10.3760/cma.j.issn.0366-6999.2012.21.018)]

102. Wang CS, Wen DX, Li SH, Li Q, Sun LY. Study on ultrasound locating interspinous space of combined spinal and epidural anesthesia in cesarean section. *Journal of Shanghai Jiaotong University (Medical Science)* 2014;**34**(11):1610-3. [DOI: [10.3969/j.issn.1674-8115.2014.11.010](https://doi.org/10.3969/j.issn.1674-8115.2014.11.010)]

103. Wu JP, Tang YZ, He LL, Zhao WX, An JX, Ni JX. Preprocedure ultrasound imaging combined with palpation technique in epidural labor analgesia. *World Journal of Clinical Cases* 2021;**9**(21):5900-8. [DOI: [10.12998/wjcc.v9.i21.5900](https://doi.org/10.12998/wjcc.v9.i21.5900)]

104. Tiourine M, Dixon AJ, Mauldin FW Jr, Scalzo D, Krishnaraj A. Imaging performance of a handheld ultrasound system with real-time computer-aided detection of lumbar spine anatomy: a feasibility study. *Investigative Radiology* 2017;**52**(8):447-55.

105. Onwochei D, Nair G, Young B, Desai N. Conventional landmark palpation versus preprocedural ultrasound for neuraxial procedures in nonobstetric patients: a systematic review with meta-analysis and trial sequential analysis of randomised controlled trials. *European Journal of Anaesthesiology* 2021;**38**(Suppl 2):S73-86.

106. Young B, Onwochei D, Desai N. Conventional landmark palpation vs. preprocedural ultrasound for neuraxial analgesia and anaesthesia in obstetrics – a systematic review and meta-analysis with trial sequential analyses. *Anaesthesia* 2021;**76**(6):818-31.

ADDITIONAL TABLES

Table 1. Overview of included studies and syntheses

Study ID ^a	Study design	Country	N randomised	Population (mean age (SD); % female)	Types of ultrasound ^b	Outcomes with available
-----------------------	--------------	---------	--------------	--------------------------------------	----------------------------------	-------------------------

Table 1. Overview of included studies and syntheses (Continued)

						data (synthesis method)^c
Abdel-hamid 2013	Parallel group RCT	Egypt	90	USG 34.7 (3.1) years; 40.0% LM 35.1 (3.3) years; 57.8%	Preprocedure	1 to 4 (all MA)
Ansari 2014	Parallel group RCT	UAE	150	USG 33.8 (4.7) years; 100% LM 34.0 (4.7) years; 100%	Preprocedure	1 to 5 (all MA)
Arzola 2015	Parallel group RCT	Canada	128	USG 32.3 (5.8) years; 100% LM 32.7 (4.7) years; 100%	Preprocedure	1 to 5, 7 (all MA except 7)
Arzola 2022	Parallel group RCT	Canada	47	USG 58.1 (15) years; 48.0% LM 58.2 (11.3) years; 88.0%	Preprocedure	4, 5 (all MA)
Auyong 2017	Parallel group RCT	USA	70	USG 58.4 (15.6) years; 45.5% LM 58.3 (16.9) years; 51.4%	Preprocedure	1, 2, 7 (all MA except 7)
Bal 2019	Parallel group RCT	India	74	USG 57.3 (6.37) years; 40.5% LM 61.1 (9.8) years; 27.0%	Preprocedure	1, 4, 5, 7 (all MA except 7)
Balaban 2017	Parallel group RCT	Turkey	40	USG 25.2 (3.0) years; 100% LM 26.1 (4.2) years; 100%	Preprocedure	1, 2, 4, 7 (all MA except 7)
Bhardwaj 2022	Parallel group RCT, three arms	India	150	USG-preprocedure 42.9 (12.7) years; 30.0% USG-real-time	Preprocedure, real-time	1 to 4, 7 (all MA)

Table 1. Overview of included studies and syntheses *(Continued)*

				43.6 (15.2) years; 28.0%		
				LM		
				39.7 (13.3) years; 24.0%		
Chin 2011	Parallel group RCT	Canada	120	USG	Preproce- dure	1 to 7 (all MA)
				62.5 (9.7) years; 66.7%		
				LM		
				61.2 (9.8) years; 63.3%		
Chin 2018	Parallel group RCT	Australia	215	USG	Preproce- dure	3 to 7 (all MA except 7)
				Not specified; 100%		
				LM		
				Not specified; 100%		
Creaney 2016	Parallel group RCT	Ireland	20	USG	Preproce- dure	1 to 3, 5 (all MA)
				29.7 (2.1) years; 100%		
				LM		
				33.2 (1.7) years; 100%		
Dhanger 2018	Parallel group RCT	India	100	USG	Preproce- dure	1, 2, 4, 7 (all MA except 7)
				23.1 (3.0) years; 100%		
				LM		
				24.0 (3.4) years; 100%		
Ekinci 2017	Parallel group RCT	Turkey	64	USG	Preproce- dure	1, 2, 4 (all MA)
				30.3 (6.6) years; 100%		
				LM		
				30.4 (7.0) years; 100%		
Elsharkawy 2017	Parallel group RCT	USA	32	USG	Real-time	1 to 3, 5, 7 (all MA except 7)
				69(10) years; 71.4%		
				LM		
				70 (10) years; 50.0%		
Galal 2019	Parallel group RCT	Egypt	178	USG	Real-time	1, 7 (all MA ex- cept 7)
				Not specified		
				LM		
				Not specified		
Gayathri 2021	Parallel group RCT	India	60	USG	Preproce- dure	1 to 3 (all MA)
				27.5 (4.0) years; 100%		

Table 1. Overview of included studies and syntheses *(Continued)*

				LM		
				26.1 (3.4) years; 100%		
Geng 2016	Parallel group RCT	China	191	USG	Preproce- dure	1, 2, 4, 5, 7 (all MA except 7)
				73(6) years; 51.0%		
				LM		
				73 (6) years; 52.0%		
Gorle 2023	Parallel group RCT	India	150	USG	Preproce- dure	1 to 4, 6, 7 (all MA except 7)
				51.7 (15.2) years; 45.3%		
				LM		
				51.7 (15.2) years; 36.0%		
Grau 2001a	Parallel group RCT	Germany	72	USG	Preproce- dure	1, 2, 5, 7 (all MA except 7)
				30.5 (5.6) years; 100%		
				LM		
				30.8 (4.8) years; 100%		
Grau 2001b	Parallel group RCT	Germany	80	USG	Preproce- dure	1, 2, 4 (all MA)
				31 (4.6) years; 100%		
				LM		
				31(3.9) years; 100%		
Grau 2002	Parallel group RCT	Germany	300	USG	Preproce- dure	1 to 3, 5, 7 (all MA)
				31.0 (4.9) years; 100%		
				LM		
				30.4 (4.9) years; 100%		
Grau 2004	Parallel group RCT, three arms	Germany	30	USG-preprocedure	Preproce- dure, re- al-time	1, 2, 4, 7 (all MA except 7)
				30.1 (4.2) years; 100%		
				USG-real-time		
				31.8 (5.1) years; 100%		
				LM		
				31.0 (5.1) years; 24.0%		
Hasanin 2017	Parallel group RCT	Egypt	48	USG	Preproce- dure	1, 2, 4, 7 (all MA)
				40 (3.3) years; 29.0%		
				LM		
				38 (4.5) years; 37.0%		

Table 1. Overview of included studies and syntheses *(Continued)*

Isiker 2023	Parallel group RCT	Turkey	59	USG 74.6 (7.4) years; 30.0% LM 75.6 (7.5) years; 37.9%	Preproce- dure	4, 5, 7 (all MA except 7)
Jagadish 2021	Parallel group RCT	India	80	USG 43.6 (15.6) years; 40.0% LM 43.6 (15.9) years; 32.5%	Preproce- dure	1, 2, 4, 7 (all MA except 7)
Jain 2019	Parallel group RCT	India	210	USG 64.7 (7.5) years; not specified LM 66.3 (7.8) years; not specified	Preproce- dure	1, 4, 5, 7 (all MA except 7)
Jayanth 2023	Parallel group RCT	India	62	USG Not specified; 100% LM Not specified; 100%	Preproce- dure	1, 4, 7 (all MA except 7)
Jiang 2021	Parallel group RCT	China	60	USG 29.2 (3.1) years; 100% LM 28.4 (3.4) years; 100%	Real-time	1 to 4, 7 (all MA except 7)
Kampitak 2018	Parallel group RCT	Thailand	70	USG 75.1 (7.0) years; 85.7% LM 73.9 (7.7) years; 91.4%	Preproce- dure	1 to 4, 6, 7 (all MA except 7)
Kawaguchi 2011	Parallel group RCT	Japan	24	USG 63 (10) years; 66.7% LM 60 (11) years; 66.7%	Preproce- dure	5, 7 (all MA except 7)
Khan 2022	Parallel group RCT	India	100	USG 54.5 (12.8) years; not specified LM 57.7 (13.2) years; not specified	Preproce- dure	1, 2, 4, 7 (all MA)

Table 1. Overview of included studies and syntheses (Continued)

Korda 2020	Parallel group RCT	South Africa	36	USG 32.1 (range 23-42) years; 100% LM 31.4 (range 22-45) years; 100%	Preproce- dure	1, 2, 4, 5 (all MA)
Kumar 2020	Parallel group RCT	India	110	USG 38.9 (13.8) years; 29.1% LM 39.0 (14.6) years; 30.9%	Preproce- dure	1 to 4, 6, 7 (all MA)
Li 2019	Parallel group RCT	China	80	USG 29.5 (3.9) years; 100% LM 30.1 (4.5) years; 100%	Preproce- dure	1 to 5, 7 (all MA)
Lim 2012	Parallel group RCT	Singapore	170	USG 61.1 (13.3) years; 35.0% LM 63.7 (12.6) years; 53.0%	Preproce- dure	1 to 5, 7 (all MA except 7)
Narkhede 2019	Parallel group RCT	India	60	USG Not specified LM Not specified	Preproce- dure	1 to 4, 7 (all MA)
Nassar 2015	Parallel group RCT	Egypt	110	USG 29.9 (5.1) years; 100% LM 28.3 (4.1) years; 100%	Preproce- dure	1, 2, 4, 7 (all MA except 7)
NCT02207972	Parallel group RCT	USA	46	USG 31.8 (range 19-39) years; 100% LM 31.8 (range 19-39) years; 100%	Preproce- dure	4, 5 (all MA)
NCT02553746	Parallel group RCT	Greece	146	USG 52.4 (18.9) years; 28.8% LM 56.3 (20.0) years; 37.9%	Preproce- dure	1 to 5, 7 (all MA except 7)

Table 1. Overview of included studies and syntheses (Continued)

Pakpirom 2022	Parallel group RCT	China	96	USG 60.0 (IQR 51-67) years; 45.8% LM 58.5 (IQR 53.75-70.25) years; 41.7%	Real-time	1 to 5, 7 (all MA)
Park 2019	Parallel group RCT	Korea	80	USG 71.1 (7.2) years; 77.5% LM 71.2 (6.1) years; 85.0%	Preproce- dure	1 to 7 (all MA except 7)
Park 2020	Parallel group RCT	Korea	44	USG 70.5 (8.8) years; 90.9% LM 66.5 (13.2) years; 77.3%	Preproce- dure	1 to 7 (all MA)
Perna 2017	Parallel group RCT	Italy	60	USG 30 (4) years; 100% LM 31 (5) years; 100%	Preproce- dure	1, 5, 7 (all MA)
Pramodrao 2022	Parallel group RCT	India	128	USG Not specified LM Not specified	Preproce- dure	1, 2, 4, 6 (all MA)
Qu 2020	Parallel group RCT	China	80	USG 83.3 (6.7) years; 75.0% LM 82.3 (7.1) years; 82.5%	Preproce- dure	1 to 5, 7 (all MA except 7)
Rahimzadeh 2022	Parallel group RCT	India	57	USG 47.4 (14.3) years; 20.7% LM 40.3 (16.0) years; 28.6%	Preproce- dure	2, 4, 5 (all MA)
Ray 2022	Parallel group RCT	India	60	USG 64.9 (5.5) years; not specified LM 67.5 (5.4) years; not specified	Preproce- dure	1 to 4, 6 (all MA)

Table 1. Overview of included studies and syntheses (Continued)

Rizk 2019	Parallel group RCT, three arms	Lebanon	180	USG-median 72.4 (7.8) years; 17.0% USG-paramedian 71.4 (7.5) years; 18.0% LM 73.5 (8.0) years; 22.0%	Preproce- dure	1 to 7 (all MA)
Sahin 2014	Parallel group RCT, four arms	Turkey	100	USG-obese 32.4 (6.6) years; 100% USG-nonobese 31.0 (4.8) years; 100% LM-obese 31.6 (5.2) years; 100% LM-nonobese 30.2 (4.7) years; 100%	Preproce- dure	1, 2, 4, 5, 7 (all MA except 7)
Singh 2023	Parallel group RCT	India	80	USG 58.5 (IQR 50.25-65.75) years; 70.0% LM 59.5 (IQR 52.25-65.75) years; 62.5%	Preproce- dure	1, 2, 5 (all MA)
Srinivasan 2015	Parallel group RCT	Ireland	100	USG 63.4 (14.1) years; 60.0% LM 65.2 (11.4) years; 48.0%	Preproce- dure	1 to 7 (all MA except 7)
Srinivasan 2018	Parallel group RCT	Ireland	119	USG 65.3 (9.7) years; 52.5% LM 68.2 (10.3) years; 50.0%	Preproce- dure	1 to 7 (all MA except 7)
Tao 2021	Parallel group RCT	China	128	USG 32.3 (5.2) years; 100% LM 30.6 (3.8) years; 100%	Preproce- dure	2, 4, 5, 7 (all MA)
Tawfik 2017	Parallel group RCT	Egypt	108	USG 27.7 (4.0) years; 100% LM	Preproce- dure	1 to 5, 7 (all MA except 7)

Table 1. Overview of included studies and syntheses (Continued)

26.7 (3.8) years; 100%						
Tawfik 2017	Parallel group RCT	Egypt	280	USG 30.2 (5.9) years; 100% LM 30.3 (5.5) years; 100%	Preproce- dure	1 to 5, 7 (all MA except 7)
Tubinis 2019	Parallel group RCT	USA	150	USG Not specified; 100% LM Not specified; 100%	Preproce- dure	1, 2, 5 (all MA)
Turkstra 2017	Parallel group RCT	Canada	80	USG 33(4) years; 100% LM 32 (5) years; 100%	Preproce- dure	1, 2, 5, 7 (all MA except 7)
Urfalioglu 2017	Parallel group RCT	Turkey	97	USG 30.5 (5.5) years; 100% LM 30.4 (5.7) years; 100%	Preproce- dure	1, 2, 7 (all MA except 7)
Uyel 2021	Parallel group RCT	Turkey	156	USG 70.2 (5.3) years; 20.5% LM 71.7 (6.5) years; 30.8%	Preproce- dure	1 to 7 (all MA)
Vallejo 2010	Parallel group RCT	USA	370	USG 27.9 (5.9) years; 100% LM 28.5 (6.1) years; 100%	Preproce- dure	1, 5, 7 (all MA except 7)
Vernon 2020	Parallel group RCT	USA	16	USG 25.3 (4.7) years; 100% LM 29.9 (5.5) years; 100%	Preproce- dure	1 to 5, 7 (all MA except 7)
Wang 2012	Parallel group RCT	China	60	USG 29.1 (4.2) years; 100% LM	Preproce- dure	1, 2, 4, 7 (all MA)

Table 1. Overview of included studies and syntheses (Continued)

29.7 (3.5) years; 100%						
Wang 2014	Parallel group RCT	China	240	USG Not specified; 100% LM Not specified; 100%	Preproce- dure	1, 2, 4, 7 (all MA except 7)
Wilkes 2017	Parallel group RCT	USA	50	USG 26.0 (5.4) years; 100% LM 26.3 (6.3) years; 100%	Preproce- dure	7 (not MA)
Wu 2021	Parallel group RCT	China	72	USG 29.3 (3.1) years; 100% LM 29.4 (2.7) years; 100%	Preproce- dure	1, 2, 4, 5, 7 (all MA except 7)

IQR: interquartile range; **LM:** landmark; **MA:** meta-analysis; **RCT:** randomised controlled trial; **USG:** ultrasound guidance

^aSee [Supplementary material 2](#) for full study details, including study inclusion, exclusion criteria and full baseline characteristics.

^bFull intervention details for each study available in [Supplementary material 2](#).

^cOutcomes shown below

1. Number of attempts until pass success or procedure termination
2. Procedure time
3. Participant satisfaction during the procedure
4. First-attempt success
5. Technical failure
6. Pain during the procedure
7. Adverse events

Table 2. Adverse events (overall and objectively assessed)

Study ID	Number of participants		Overall		Unintentional dural puncture		Bloody tap or bloody puncture	
	USG	LM	USG	LM	USG	LM	USG	LM
Arzola 2015	60	68	NS	NS	1 (1.7)	0	NS	NS
Bal 2019	37	37	NS	NS	NS	NS	4 (10.8)	2 (5.4)
Balaban 2017	20	20	NS	NS	0	0	1 (5.0)	2 (10.0)
Bhardwaj 2022	P 50; R 50	50	P 1 (2.0); R 1 (2.0)	2 (4.0)	NS	NS	NS	NS
Chin 2011	60	60	0	0	NS	NS	NS	NS
Chin 2018	105	110	NS	NS	2 (1.9)	3 (2.7)	NS	NS
Dhanger 2018	50	50	NS	NS	NS	NS	0	7 (14.0)
Elsharkawy 2017	14	18	NS	NS	NS	NS	NS	NS
Galal 2019	89	89	NS	NS	NS	NS	8 (9.0)	37 (41.6)
Geng 2016	98	93	NS	NS	NS	NS	NS	NS
Gorle 2023	75	75	NS	NS	NS	NS	13 (17.3)	14 (18.7)
Grau 2001a	36	36	NS	NS	NS	NS	NS	NS
Grau 2002	150	150	33 (22.0)	54 (36.0)	1 (0.7)	2 (1.3)	3 (2.0)	11 (7.3)
Grau 2004	P 10; R 10	10	NS	NS	NS	NS	NS	NS
Hasanin 2017	24	24	1 (4.2)	4 (16.7)	0	1 (4.2)	1 (4.2)	4 (16.7)
Isiker 2023	30	29	NS	NS	NS	NS	1 (3.3)	5 (17.2)
Jagadish 2021	40	40	NS	NS	0	1 (2.5)	NS	NS
Jain 2019	105	105	NS	NS	0	6 (5.7)	NS	NS

Table 2. Adverse events (overall and objectively assessed) *(Continued)*

Jayanth 2023	31	31	NS	NS	0	2 (6.5)	NS	NS
Jiang 2021	30	30	NS	NS	0	2 (6.7)	4 (13.3)	5 (16.7)
Kampitak 2018	35	35	NS	NS	NS	NS	0	7 (20.0)
Kawaguchi 2011	12	12	NS	NS	NS	NS	NS	NS
Khan 2022	50	50	0	0	0	0	0	0
Kumar 2020	55	55	0	0	NS	NS	0	0
Li 2019	40	40	0	2 (5.0)	NS	NS	0	2 (5.0)
Lim 2012	85	85	NS	NS	NS	NS	9 (10.6)	9 (10.6)
Narkhede 2019	30	30	0	0	NS	NS	NS	NS
Nassar 2015	55	55	NS	NS	0	2 (3.6)	1 (1.8)	1 (1.8)
NCT02553746	77	64	NS	NS	NS	NS	NS	NS
Pakpirom 2022	48	48	1 (2.1)	0	NS	NS	1 (2.1)	0
Park 2019	40	40	NS	NS	NS	NS	1 (2.5)	1 (2.5)
Park 2020	22	22	2 (9.1)	4 (18.2)	NS	NS	1 (4.5)	2 (9.1)
Perna 2017	30	30	0	0	0	0	0	0
Qu 2020	40	40	NS	NS	1 (2.5)	2 (5.0)	1 (2.5)	3 (7.5)
Rizk 2019	M 60; Pa 60	60	M 9 (15.0); Pa 20 (33.3)	10 (16.7)	NS	NS	M 8 (13.3); Pa 15 (25.0)	9 (15.0)
Sahin 2014	N 25; O 25	N 25; O 25	NS	NS	NS	NS	N 6 (24.0); O 8 (32.0)	N 9 (36.0); O 7 (28.0)
Srinivasan 2015	50	50	NS	NS	NS	NS	0	2 (4.0)
Srinivasan 2018	59	60	NS	NS	NS	NS	6 (10.2)	2 (3.3)

Table 2. Adverse events (overall and objectively assessed) (Continued)

Tao 2021	64	64	2 (3.1)	3 (4.7)	0	0	NS	NS
Tawfik 2017	53	55	NS	NS	1 (1.9)	0	2 (3.8)	3 (5.5)
Tawfik 2023	140	140	NS	NS	NS	NS	9 (6.4)	9 (6.4)
Turkstra 2017	40	40	NS	NS	NS	NS	3 (7.5)	3 (7.5)
Urfalioglu 2017	48	49	NS	NS	NS	NS	NS	NS
Uyel 2021	78	78	0	0	NS	NS	NS	NS
Vallejo 2010	189	181	NS	NS	1 (0.5)	1 (0.6)	NS	NS
Vernon 2020	7	9	NS	NS	0	1 (11.1)	NS	NS
Wang 2012	30	30	2 (6.7)	6 (20.0)	0	0	2 (6.7)	6 (20.0)
Wang 2014	120	120	NS	NS	NS	NS	1 (0.8)	8 (6.7)
Wilkes 2017	22	28	NS	NS	NS	NS	NS	NS
Wu 2021	36	36	NS	NS	1 (2.8)	1 (2.8)	1 (2.8)	6 (16.7)

USG: ultrasound guidance, **LM:** landmark method, **NS:** not specified, **P:** preprocedure, **R:** real-time, **M:** midline, **Pa:** paramedian, **N:** normal, **O:** obese

Values are n (%)

Severe complications were reported by three studies (Chin 2018; Geng 2016; Gorle 2023). Two cases were reported only in the LM group of Gorle 2023. No cases were reported in the other groups.

Spinal haematoma was stated only in Narkhede 2019. No cases were reported in either the USG or LM groups.

Infection of the puncture site or spine was not stated in any studies.

Table 3. Adverse events (participant-reported outcomes)

Study ID	Number of participants		Back pain		Headache		Radicular pain or paraesthesia	
	USG	LM	USG	LM	USG	LM	USG	LM
Arzola 2015	60	68	NS	NS	NS	NS	NS	NS
Bal 2019	37	37	NS	NS	NS	NS	NS	NS

Table 3. Adverse events (participant-reported outcomes) *(Continued)*

Balaban 2017	20	20	0	3 (15.0)	NS	NS	2 (10.0)	4 (20.0)
Bhardwaj 2022	P 50; R 50	50	NS	NS	NS	NS	P 1 (2.0); R 1 (2.0)	2 (4.0)
Chin 2011	60	60	NS	NS	NS	NS	NS	NS
Chin 2018	105	110	NS	NS	NS	NS	NS	NS
Dhanger 2018	50	50	NS	NS	NS	NS	1 (2.0)	14 (28.0)
Elsharkawy 2017	14	18	NS	NS	0	1 (5.6)	NS	NS
Galal 2019	89	89	NS	NS	NS	NS	9 (10.1)	35 (39.3)
Geng 2016	98	93	NS	NS	0	0	2 (2.0)	2 (2.2)
Gorle 2023	75	75	NS	NS	NS	NS	0	2 (2.7)
Grau 2001a	36	36	7 (19.4)	7 (19.4)	2 (5.6)	4 (11.1)	NS	NS
Grau 2002	150	150	22 (14.7)	33 (22.0)	7 (4.7)	28 (18.7)	2 (1.3) (2.0)	3 (2.0)
Grau 2004	P 10; R 10	10 (100.0)	P 1 (10.0); R 0	3 (30.0)	P 1 (10.0); R 0	2 (20.0)	NS	NS
Hasanin 2017	24	24	NS	NS	NS	NS	NS	NS
Isiker 2023	30	29	NS	NS	NS	NS	3 (10.0)	2 (6.9)
Jagadish 2021	40	40	NS	NS	NS	NS	NS	NS
Jain 2019	105	105	NS	NS	NS	NS	NS	NS
Jayanth 2023	31	31	NS	NS	NS	NS	NS	NS
Jiang 2021	30	30	0	1 (3.3)	NS	NS	1 (3.3)	6 (20.0)
Kampitak 2018	35	35	NS	NS	NS	NS	0	6 (17.1)
Kawaguchi 2011 ‡	12	12	NS	NS	NS	NS	NS	NS

Table 3. Adverse events (participant-reported outcomes) *(Continued)*

Khan 2022	50	50	NS	NS	NS	NS	0	0
Kumar 2020	55	55	NS	NS	NS	NS	0	0
Li 2019	40	40	0	0	0	0	0	0
Lim 2012	85	85	NS	NS	NS	NS	11 (12.9)	4 (4.7)
Narkhede 2019	30	30	NS	NS	0	0	0	0
Nassar 2015	55	55	NS	NS	NS	NS	NS	NS
NCT02553746	77	64	11 (14.3)	11 (17.2)	NS	NS	NS	NS
Pakpirom 2022	48	48	NS	NS	NS	NS	NS	NS
Park 2019	40	40	NS	NS	NS	NS	2 (5.0)	2 (5.0)
Park 2020	22	22	NS	NS	NS	NS	1 (4.5)	4 (18.2)
Perna 2017	30	30	NS	NS	NS	NS	NS	NS
Qu 2020	40	40	0	0	0	0	2 (5.0)	2 (5.0)
Rizk 2019	M 60; Pa 60	60	NS	NS	NS	NS	M 1 (1.7); Pa 11 (18.3)	1 (1.7)
Sahin 2014	N 25; O 25	N 25; O 25	N 0; O 0	N 1 (4.0); O 3 (12.0)	N 2 (8.0); O 0	N 0; O 1 (4.0)	N 4 (16.0); O 1 (4.0)	N 2 (8.0); O 6 (24.0)
Srinivasan 2015	50	50	NS	NS	NS	NS	3 (6.0)	2 (4.0)
Srinivasan 2018	59	60	NS	NS	NS	NS	11 (20.8)	5 (8.3)
Tao 2021	64	64	2 (3.1)	3 (4.7)	NS	NS	0	0
Tawfik 2017	53	55	9 (17.0)	7 (12.7)	NS	NS	NS	NS
Tawfik 2023	140	140	NS	NS	NS	NS	16 (11.4)	17 (12.1)
Turkstra 2017	40	40	NS	NS	NS	NS	3 (7.5)	1 (2.5)

Table 3. Adverse events (participant-reported outcomes) (Continued)

Urfalioglu 2017	48	49	7 (14.6)	21 (42.9)	7 (14.6)	5 (10.2)	NS	NS
Uyel 2021	78	78	NS	NS	NS	NS	NS	NS
Vallejo 2010	189	181	NS	NS	2 (1.1)	0	NS	NS
Vernon 2020	7	9	NS	NS	NS	NS	0	2 (22.2)
Wang 2012	30	30	0	0	0	0	0	0
Wang 2014	120	120	10 (8.3)	22 (18.3)	NS	NS	4 (3.3)	10 (8.3)
Wilkes 2017	22	28	4 (18.2)	7 (25.0)	NS	NS	NS	NS
Wu 2021	36	36	NS	NS	NS	NS	1 (2.8)	4 (11.1)
USG : ultrasound guidance, LM : landmark method, NS : not specified, P : preprocedure, R : real-time, M : midline, Pa : paramedian, N : normal, O : obese								

Values are n (%)

‡ Kawaguchi 2011 reported on postoperative nausea and vomiting (PONV) as follows: "PONV was assessed using scores (0, none; 1, mild; 2, discomforting; and 3, severe or requiring rescue medication). The PONV score in the USG group was significantly lower [0 (0–0) vs. 1 (0–3), $P=0.03$]. Data were expressed as median (interquartile range)."

INDEX TERMS

Medical Subject Headings (MeSH)

*Anatomic Landmarks; *Anesthesia, Epidural [adverse effects] [methods]; *Anesthesia, Spinal [adverse effects] [methods]; Bias; Patient Satisfaction; Randomized Controlled Trials as Topic; *Ultrasonography, Interventional [methods]

MeSH check words

Adult; Female; Humans